Lecture 1: Randomness & Confidence

Probability, Concentration, and Markov Dynamics

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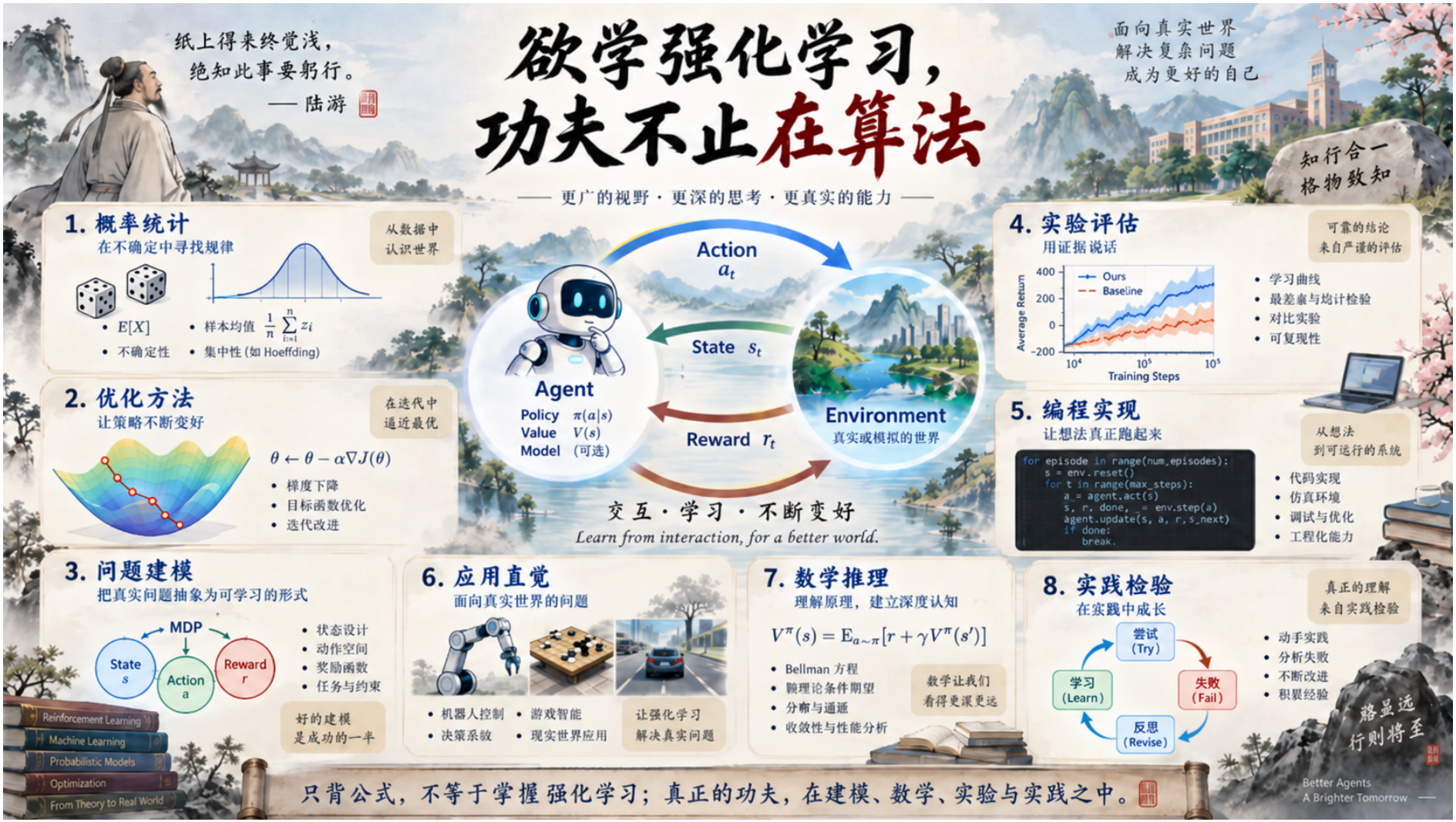
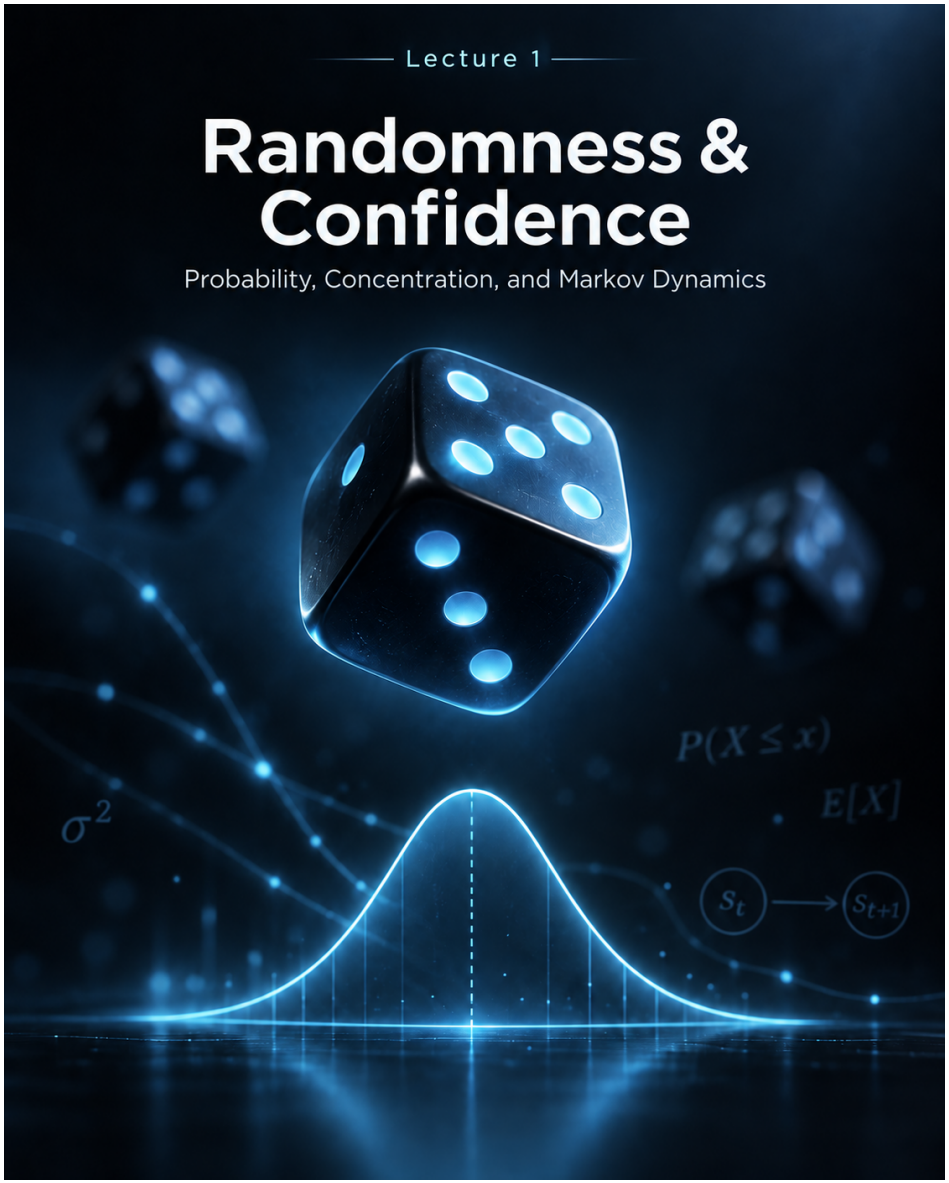
Reinforcement Learning

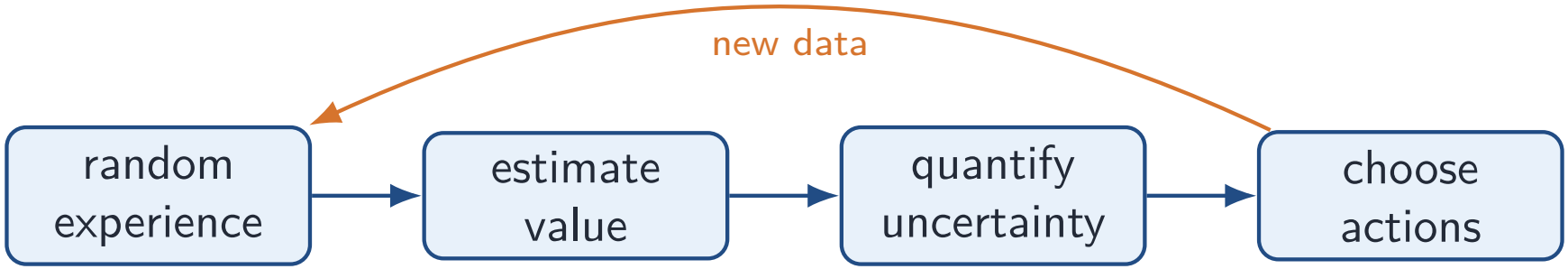
汝果欲学诗，
功夫在诗外。
陆游
游游
—— 学诗不止于辞章，更在学问、阅历、见识与胸怀。 ——
博览群书
读万卷书
开阔眼界
庚癸全书
史记
体察人生
观人间百态
知世事冷暖
亲近山水
观山川草木
感自然之美
纸上得来终觉浅
绝知此事要躬行
——陆游
感受四时
春花秋月冬雪
体会生命流转
行走四方
见不同风土
增广见识
行万里路
诗在心中
而不止于纸上
心系家国
关怀社会天下
涵养胸怀
读诗心
连着大千世界


纸上得来终觉浅，
绝知此事要躬行。
—— 陆游
欲学强化学习，
功夫不止在算法
—— 更广的视野 · 更深的思考 · 更真实的能力 ——
面向真实世界
解决复杂问题
成为更好的自己
知行合一
格物致知
1. 概率统计
在不确定中寻找规律
从数据中
认识世界
• $E[X]$
• 样本均值 $\frac{1}{n}\sum_{i=1}^{n} z_i$
• 不确定性
• 集中性 (如 Hoeffding)
2. 优化方法
让策略不断变好
在迭代中
逼近最优
$\theta \leftarrow \theta - \alpha \nabla J(\theta)$
• 梯度下降
• 目标函数优化
• 迭代改进
Action
a_t
State s_t
Reward r_t
Agent
Policy $\pi(a|s)$
Value $V(s)$
Model (可选)
Environment
真实或模拟的世界
交互 · 学习 · 不断变好
Learn from interaction, for a better world.
4. 实验评估
用证据说话
可靠的结论
来自严谨的评估
Ours
Baseline
Average Return
Training Steps
• 学习曲线
• 最差值与均计检验
• 对比实验
• 可复现性
5. 编程实现
让想法真正跑起来
从想法
到可运行的系统
for episode in range(num_episodes):
s = env.reset()
for t in range(max_steps):
a = agent.act(s)
s, r, done, _ = env.step(a)
agent.update(s, a, r, s_next)
if done:
break
• 代码实现
• 仿真环境
• 调试与优化
• 工程化能力
3. 问题建模
把真实问题抽象为可学习的形式
MDP
State
Action
Reward
• 状态设计
• 动作空间
• 奖励函数
• 任务与约束
好的建模
是成功的一半
Reinforcement Learning
Machine Learning
Probabilistic Models
Optimization
From Theory to Real World
6. 应用直觉
面向真实世界的问题
• 机器人控制
• 游戏智能
• 决策系统
• 现实世界应用
让强化学习
解决真实问题
7. 数学推理
理解原理，建立深度认知
$V^{\pi}(s) = \mathrm{E}_{a\sim\pi}[r + \gamma V^{\pi}(s')]$
• Bellman 方程
• 期望条件期望
• 分布与通道
• 收敛性与性能分析
数学让我们
看得更深更远
8. 实践检验
在实践中成长
真正的理解
来自实践检验
尝试
(Try)
失败
(Fail)
反思
(Revise)
学习
(Learn)
• 动手实践
• 分析失败
• 不断改进
• 积累经验
路虽远
行则将至
只背公式，不等于掌握强化学习；真正的功夫，在建模、数学、实验与实践之中。
Better Agents
A Brighter Tomorrow

Lecture 1 at a Glance



Why This Lecture Matters for RL



Core idea

Reinforcement learning is not only about optimizing a policy; it is also about learning reliable estimates from dependent, action-generated data.

Learning Goals

By the end of Lecture 1, you should be able to:

1. use expectation and conditional expectation as value operators;
2. explain Monte Carlo estimation and why concentration matters;
3. compute stationary distributions for simple Markov chains;
4. interpret mixing as "forgetting the initial state";
5. perform a simple Bayesian update and connect it to belief states.

Outline

1. Review of Probability & Statistics
2. Expectation & Conditional Expectation
3. Conditional Expectation as Projection Geometry
4. Concentration Inequality
5. Markov Dynamics I
6. Markov Dynamics II
7. Statistical Inference
8. Bayesian Learning: RL Perspective
9. Summary

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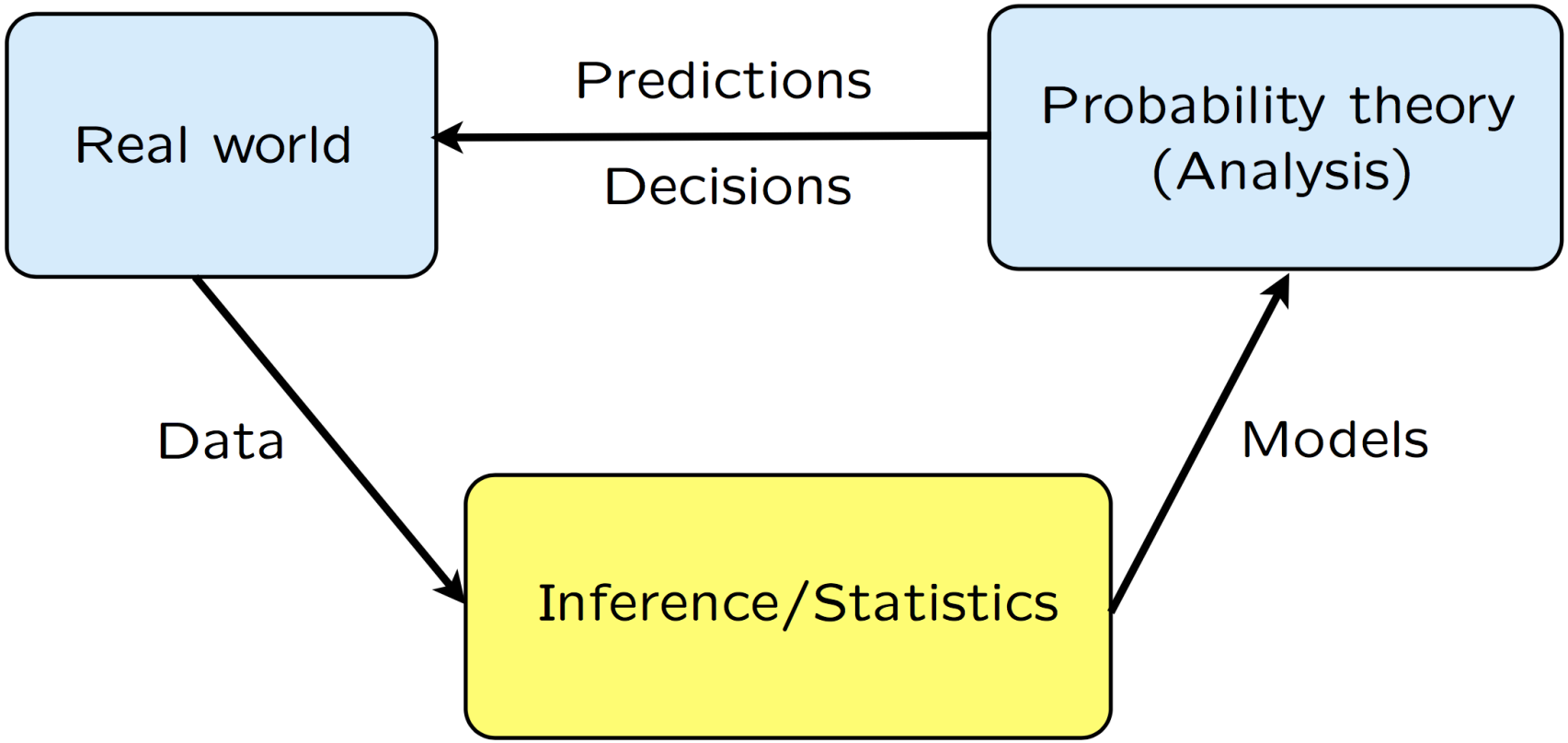
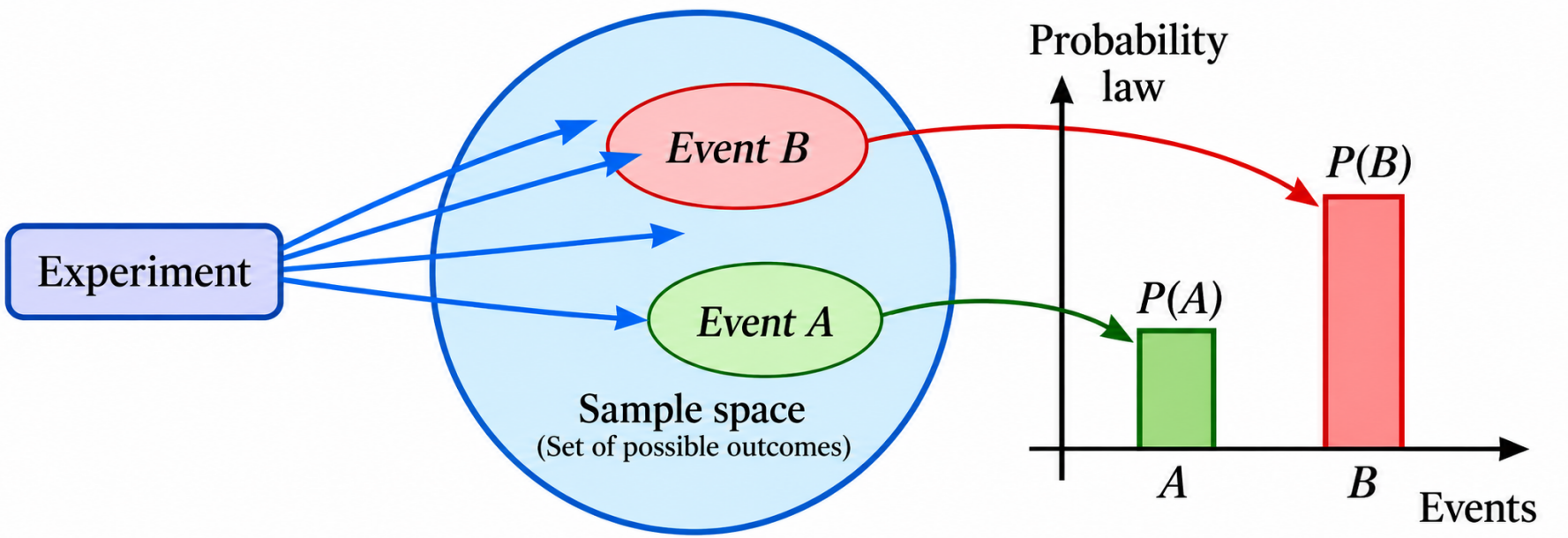
Understanding Random Worlds: Framework



Understanding Random Worlds: Modeling



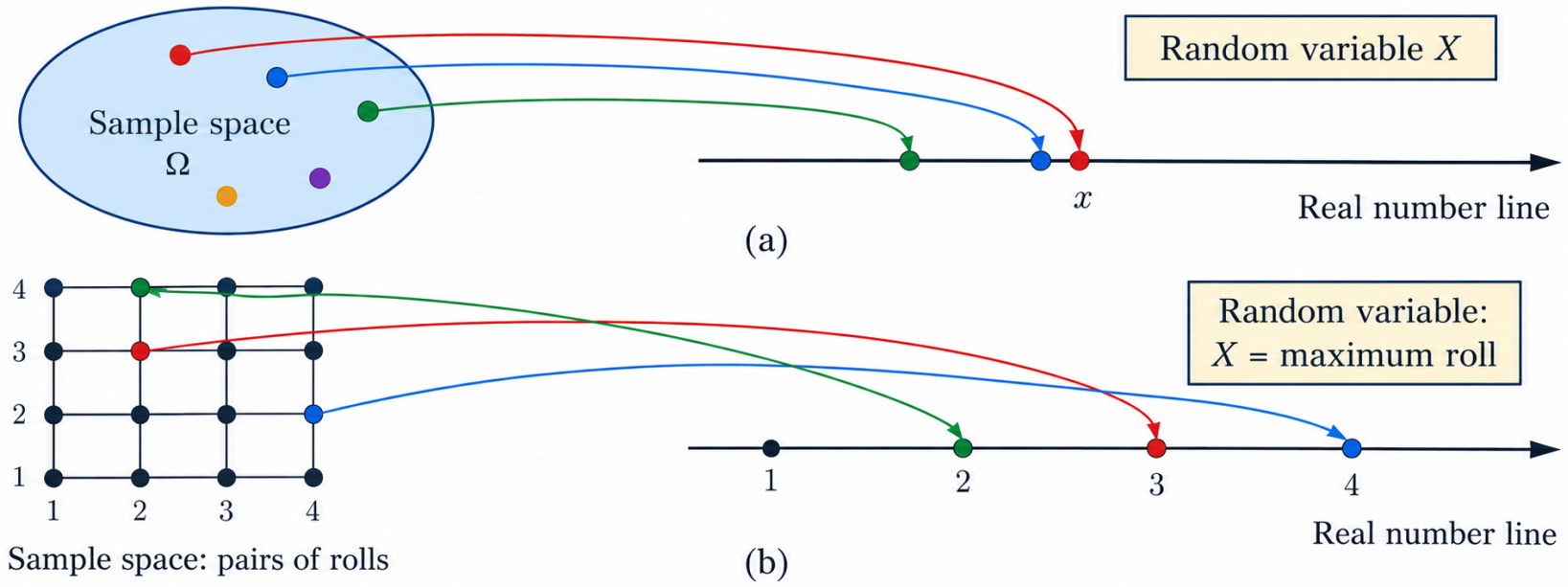
Bayes' Rule

Theorem

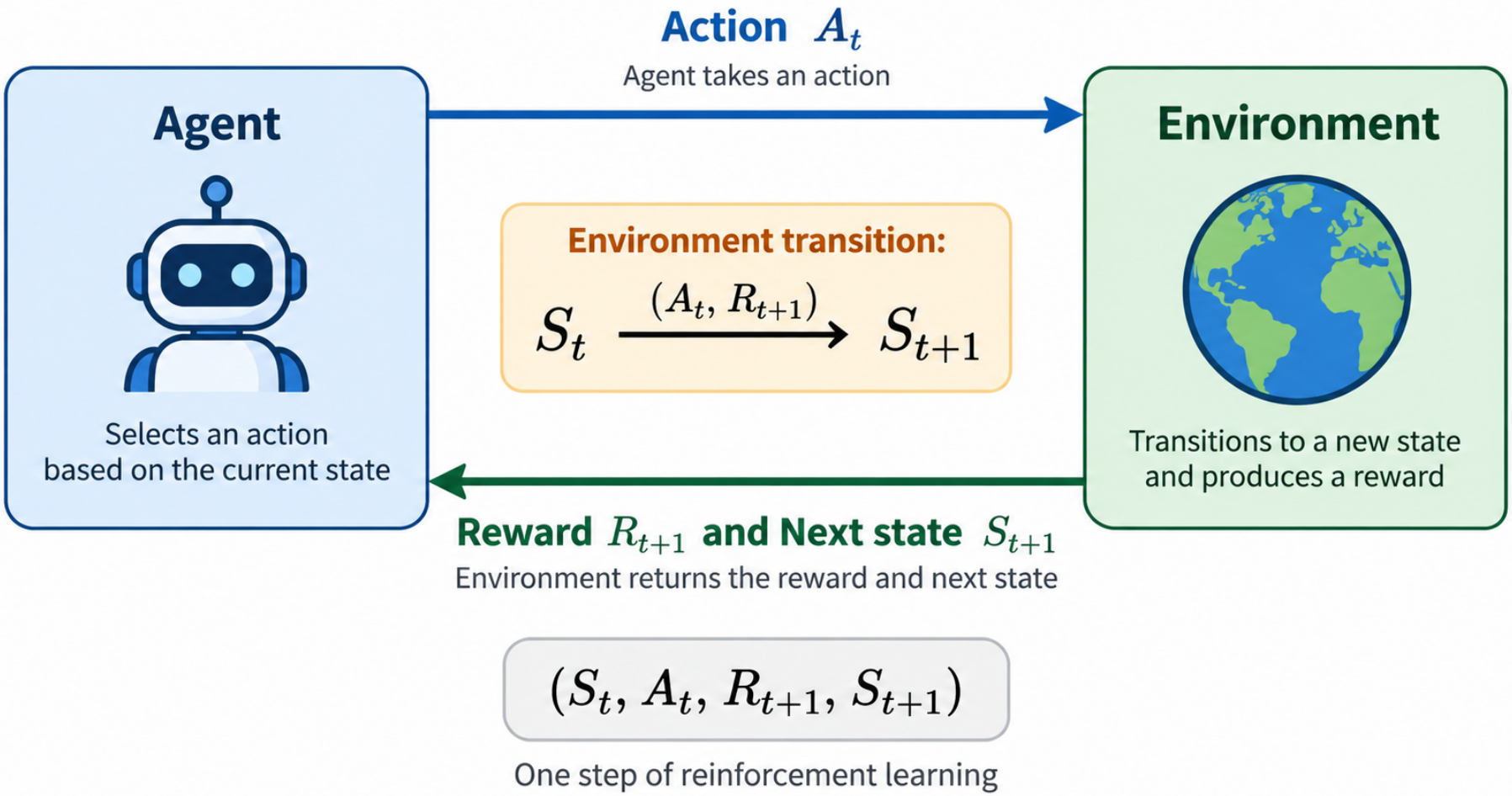
For any events A and B with positive probabilities,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Understanding Random Worlds: Modeling



Random Variables: RL Perspective



General Bayes' Rule

	Y **discrete**	Y **continuous**
X **discrete**	$P(Y=y\|X=x)=\frac{P(X=x\|Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y\|X=x)=\frac{P(X=x\|Y=y)f_Y(y)}{P(X=x)}$
X **continuous**	$P(Y=y\|X=x)=\frac{f_X(x\|Y=y)P(Y=y)}{f_X(x)}$	$f_{Y\|X}(y\|x)=\frac{f_{X\|Y}(x\|y)f_Y(y)}{f_X(x)}$

General LOTP(Law of Total Probability)

	Y **discrete**	Y **continuous**
X **discrete**	$P(X=x)=\sum_y P(X=x\|Y=y)P(Y=y)$	$P(X=x)=\int_{-\infty}^{\infty} P(X=x\|Y=y)f_Y(y)dy$
X **continuous**	$f_X(x)=\sum_y f_X(x\|Y=y)P(Y=y)$	$f_X(x)=\int_{-\infty}^{\infty} f_{X\|Y}(x\|y)f_Y(y)dy$

Probability is a Belief

WE UPDATE OUR BELIEFS WHEN WE SEE EVIDENCE

belief + evidence → updated belief

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Expectation: RL Perspective

Reinforcement learning is based on the reward hypothesis:

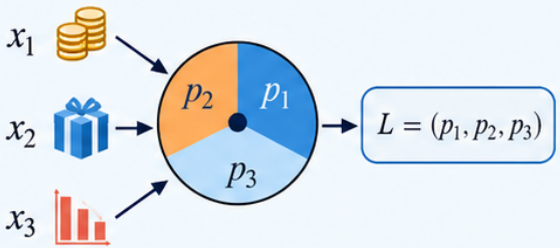
Theorem

All goals can be described by the maximization of expected cumulative reward

Von Neumann–Morgenstern Expected Utility

Expected Utility Theorem

Lotteries



$$U(L) = \sum_i p_i u(x_i)$$

Axioms

- **Completeness**
- **Transitivity**
- **Continuity**
- **Independence**

Expected Utility Theorem

Representation

$$L \succeq M \iff \sum_i p_i u(x_i) \geq \sum_i q_i u(x_i)$$

$u \sim au + b, \; a > 0$

positive affine

Risk attitude / RL connection

safe

$x = 0$

$(p = 1)$

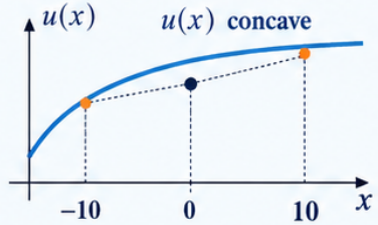
$\mathbb{E}[X] = 0$

risky

$x = +10 \quad (p = 0.5)$

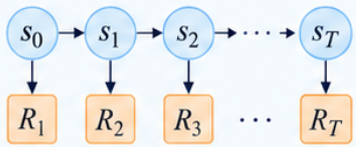
$x = -10 \quad (p = 0.5)$

$\mathbb{E}[X] = 0$



$\mathbb{E}[u(X)]_{risky} < u(\mathbb{E}[X]) = \mathbb{E}[u(X)]_{safe}$

trajectory τ



extra temporal structure

$$\mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t R_{t+1}\right]$$

$\mathbb{E}[u(\tau)]$

Summary

1. Axioms $\Rightarrow$ Expected Utility
2. Lotteries ranked by $\mathbb{E}[u(X)]$
3. Cumulative reward needs extra temporal structure

Expectation as Long-Run Average

Definition

For a discrete random variable,

$$\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x).$$

For a continuous random variable,

$$\mathbb{E}[X] = \int xf(x)\,dx.$$

Two-step example

A reward is $+10$ with probability 0.3 and -2 with probability 0.7.

$$\mathbb{E}[R] = 10 \cdot 0.3 + (-2) \cdot 0.7 = 1.6.$$

Linearity of Expectation

Key fact

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

No independence is required.

Why this matters

$$\mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t R_{t+1}\right] = \sum_{t=0}^{T-1} \gamma^t \mathbb{E}[R_{t+1}].$$

Common mistake

Linearity of expectation is always true; variance linearity usually needs independence.

Conditional Expectation

- Conditional expectation is a powerful tool for calculating expectations: first-step analysis
- Conditional expectation allows us to predict or estimate unknowns based on whatever evidence is currently available.
- Conditional Expectation given an event: $\mathbb{E}(Y|A)$
- Conditional Expectation given a random variable: $\mathbb{E}(Y|X)$

Conditional Expectation Given An Event

Definition

Let A be an event with positive probability. If Y is a discrete r.v., then the *conditional expectation of Y given A* is

$$\mathbb{E}(Y|A) = \sum_y y\, P(Y = y|A),$$

where the sum is over the support of Y. If Y is a continuous r.v. with PDF f, then

$$\mathbb{E}(Y|A) = \int_{-\infty}^{\infty} y\, f(y|A)\, dy,$$

where the conditional PDF $f(y|A)$ is defined as the derivative of the conditional CDF $F(y|A) = P(Y \leq y|A)$, and can also be computed by a hybrid version of Bayes' rule:

$$f(y|A) = \frac{P(A|Y = y)\, f(y)}{P(A)}.$$

Intuition for $E(Y|A)$

Principle

$E(Y|A)$ is approximately the average of Y in a large number of simulation runs in which A occurred.

Law of Total Expectation

Theorem

Let $A_1, ..., A_n$ be a partition of a sample space, with $P(A_i) > 0$ for all i, and let Y be a random variable on this sample space. Then

$$E(Y) = \sum_{i=1}^{n} E(Y|A_i) P(A_i).$$

Conditional Expectation: RL Perspective

RL Elements

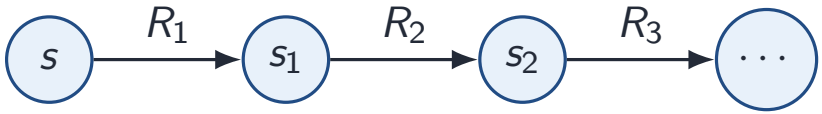
- reward: R_t
- state: S_t
- action: A_t
- return: $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$

Return

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$$

Value function

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s].$$



A value function is not a deterministic score. It is an average over many possible futures given current state.

Conditional Expectation: RL Perspective

Intuition

$\mathbb{E}[X \mid Y = y]$ is the best average prediction of X after observing $Y = y$.

Discrete formula

$$\mathbb{E}[X \mid Y = y] = \sum_x x \, \mathbb{P}(X = x \mid Y = y).$$

RL reading

$$\mathbb{E}[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s, A_t = a]$$

means: average the one-step reward plus future value after fixing the current state-action pair.

Small Example: Condition First, Average Later

A robot first observes weather $W \in \{\text{sunny}, \text{rainy}\}$.

$$\mathbb{P}(W = \text{sunny}) = 0.6, \quad \mathbb{P}(W = \text{rainy}) = 0.4.$$

Expected reward after observing weather:

$$\mathbb{E}[R \mid W = \text{sunny}] = 5, \qquad \mathbb{E}[R \mid W = \text{rainy}] = 1.$$

Compute

$$\mathbb{E}[R] = 0.6 \cdot 5 + 0.4 \cdot 1 = 3.4.$$

Small Example: Condition First, Average Later

A robot first observes weather $W \in \{\text{sunny}, \text{rainy}\}$.

$$\mathbb{P}(W = \text{sunny}) = 0.6, \quad \mathbb{P}(W = \text{rainy}) = 0.4.$$

Expected reward after observing weather:

$$\mathbb{E}[R \mid W = \text{sunny}] = 5, \qquad \mathbb{E}[R \mid W = \text{rainy}] = 1.$$

Compute

$$\mathbb{E}[R] = 0.6 \cdot 5 + 0.4 \cdot 1 = 3.4.$$

RL interpretation

The state partitions possible futures. Value is the weighted average over those futures.

Conditional Expectation Given An R.V.

Definition

Let $g(x) = E(Y|X = x)$. Then the *conditional expectation of Y given X*, denoted $E(Y|X)$, is defined to be the random variable $g(X)$. In other words, if after doing the experiment X crystallizes into x, then $E(Y|X)$ crystallizes into $g(x)$.

Remark

- $E(Y|X)$ is a function of X, and it is a random variable.
- It makes sense to computer $E(E(Y|X))$ and $Var(E(Y|X))$.

Property I: Dropping What's Independent

Theorem

If X and Y are independent, then $E(Y|X) = E(Y)$.

Property II: Taking Out What's Known

Theorem

For any function h,

$$E\left(h\left(X\right)Y|X\right)=h\left(X\right)E\left(Y|X\right)$$

Property III: Linearity

Theorem

$E(Y_1 + Y_2|X) = E(Y_1|X) + E(Y_2|X)$.

Adam's Law

Theorem (Law of Iterated Expectation)

For any r.v.s X and Y,

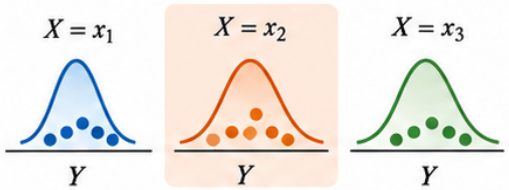
$$E(E(Y|X)) = E(Y).$$

Law of Iterated Expectation

$$E(E(Y \mid X)) = E(Y)$$

Step 1: Inner expectation

Fix $X = x$.



$$m(x) = E(Y \mid X = x)$$

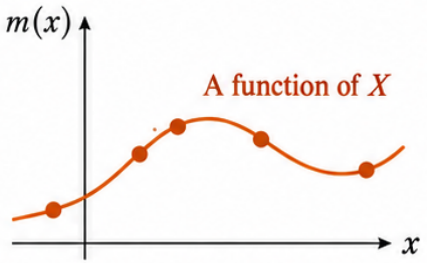
$$m(x) = \int y f_{Y|X}(y \mid x)\, dy$$

Average Y over the conditional distribution of Y given $X = x$.

Conditional mean as a function of X

$$E(Y \mid X) = m(X)$$

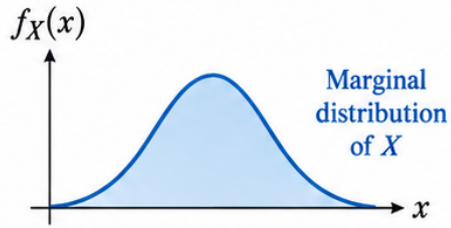
is itself a random variable depending on X.



The conditional mean $m(X)$ varies with the value of X.

Step 2: Outer expectation

Average the values $m(X)$ over the marginal distribution of X.



$$E(E(Y \mid X)) = E(m(X))$$
$$= \int m(x) f_X(x)\, dx$$

Average the conditional mean using the marginal distribution of X.

Intuition: **First average within each** group (condition on X), then take a weighted average across groups (using the distribution of X).

No independence is required. Assume $E|Y| < \infty$.

Understanding

$$
\begin{aligned}
E\big(E(Y \mid X)\big) &= \int_{-\infty}^{\infty} \underbrace{\left[\int_{-\infty}^{\infty} y\, f_{Y|X}(y \mid x)\, \mathrm{d}y\right]}_{\text{内层：按 } Y|X=x \text{ 的条件分布求平均}} \underbrace{f_X(x)\, \mathrm{d}x}_{\text{外层：按 } X \text{ 的边缘分布求平均}} \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y\, f_{X,Y}(x,y)\, \mathrm{d}y\, \mathrm{d}x \\
&= \int_{-\infty}^{\infty} y\, f_Y(y)\, \mathrm{d}y = E(Y).
\end{aligned}
$$

直观理解：先求“各组内部的平均值”，再按“各组所占比例”加权平均，就得到总体平均值。 例如，X 表示班级、Y 表示成绩，内层求各班平均成绩，外层按各班人数占比加权，而不是对各班均分做简单平均。

Adam's Law with Extra Conditioning

Theorem (Law of Iterated Expectation with Extra Conditioning)

For any r.v.s X, Y, Z, we have

$$E(E(Y|X,Z)|Z) = E(Y|Z)$$
$$E(E(X|Z,Y)|Y) = E(X|Y)$$

Conditional Variance

Definition

The *conditional variance of Y given X* is

$$\operatorname{Var}(Y|X) = E\left((Y - E(Y|X))^2 |X\right).$$

This is equivalent to

$$\operatorname{Var}(Y|X) = E\left(Y^2|X\right) - (E(Y|X))^2.$$

Eve's law

Theorem (Law of Total Variance)

For any r.v.s X and Y,

$$\operatorname{Var}(Y) = E(\operatorname{Var}(Y|X)) + \operatorname{Var}(E(Y|X)).$$

The ordering of E's and Var's on the right-hand side spells EVVE, whence the name Eve's law. Eve's law is also known as the law of total variance or the variance decomposition formula.

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Prediction & Estimation

- Estimate Y from the observed value X
- Choose the estimator (inference function) $g(\cdot)$ to minimize the expected error $E(c(Y, g(X)))$
- $c(Y, \hat{Y})$ is the cost of guessing $\hat{Y}$ when the actually value is Y.
- When $c(Y, \hat{Y}) = ||Y - \hat{Y}||^2$, the best guess is called "the least square estimate (LSE)" estimate of Y given X.
- Further, if the function $g(\cdot)$ is restricted to be linear, i.e., of the form $a + bX$, it is called "the Linear Least Square Estimate (LLSE)" estimate of Y given X.
- Further, if the function $g(\cdot)$ can be arbitrary, it is called "the Minimum Mean Square Estimate (MMSE)" estimate of Y given X.

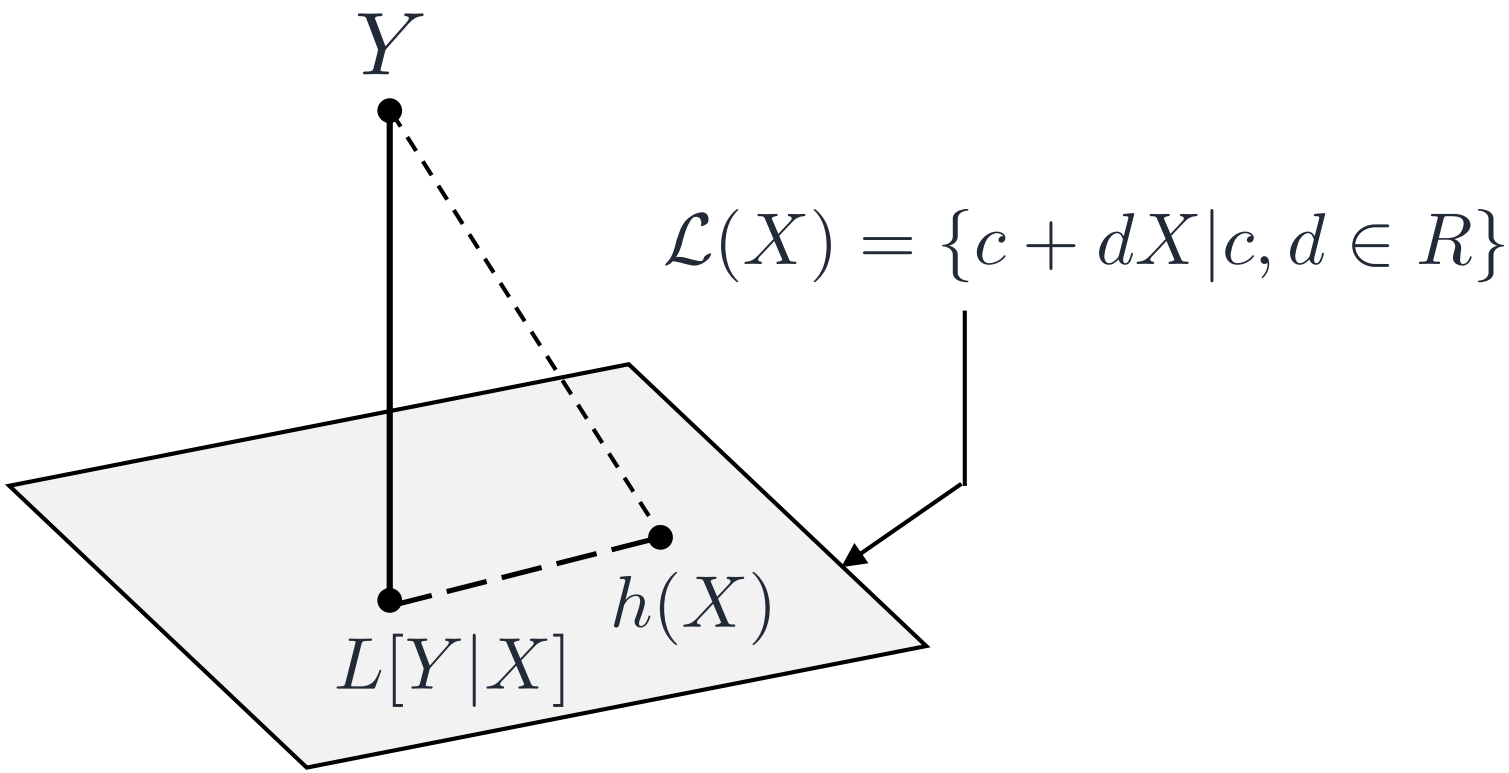
Linear Least Square Estimate

Theorem

The Linear Least Square Estimate (LLSE) of Y given X, denoted by $L[Y|X]$, is the linear function $a + bX$ that minimizes $E[(Y - a - bX)^2]$. In fact,

$$L[Y|X] = E(Y) + \frac{Cov(X, Y)}{Var(X)}(X - E(X))$$

LLSE: Projection Perspective



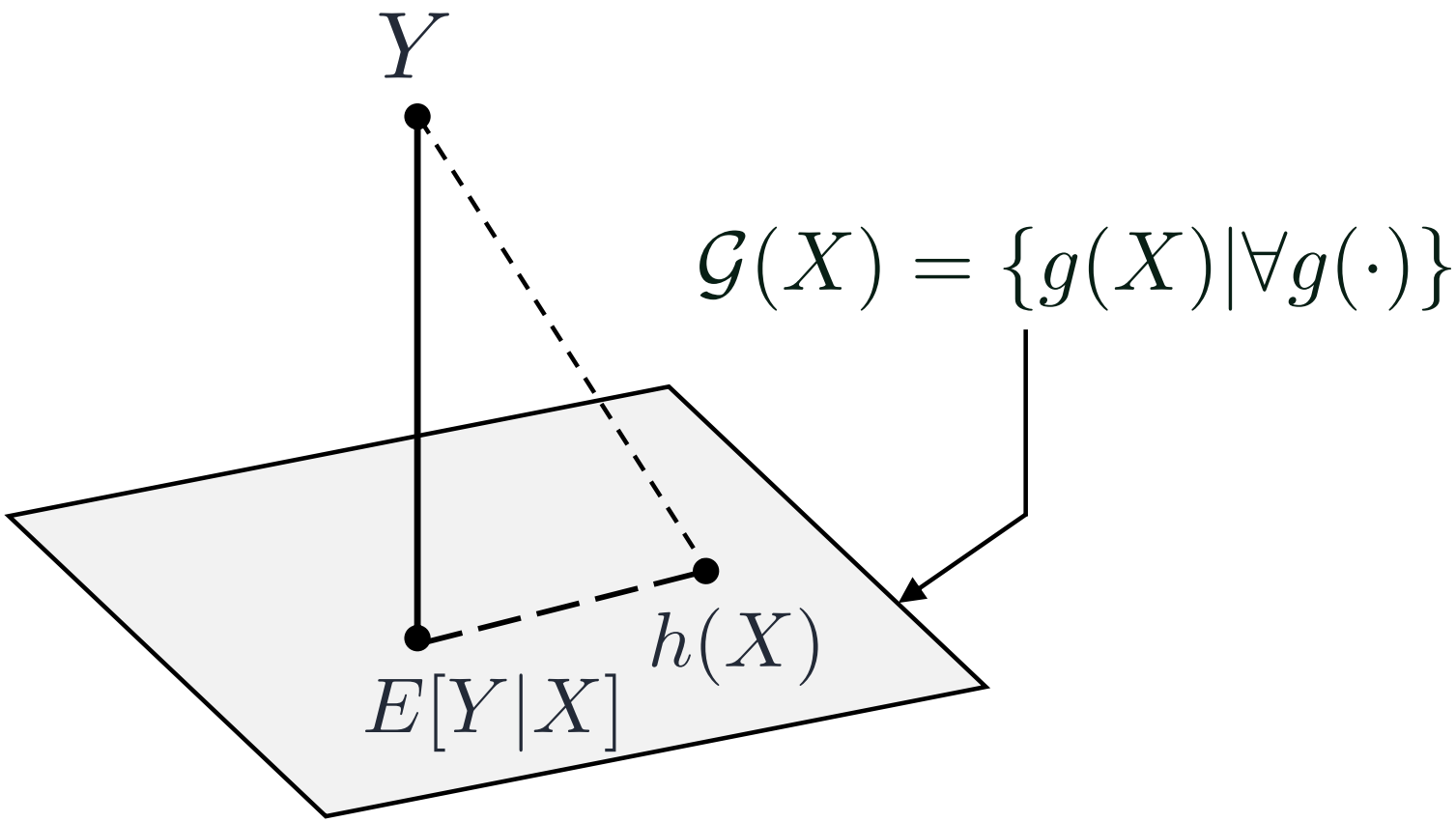
Minimum Mean Square Error Estimator

Theorem

The MMSE of Y given X is given by

$$g(X) = E[Y|X]$$

MMSE: Projection Perspective



Projection Interpretation

Theorem

For any function h, the r.v. $Y - E(Y|X)$ is uncorrelated with $h(X)$. Equivalently,

$$E((Y - E(Y|X))h(X)) = 0.$$

(This is equivalent since $E(Y - E(Y|X)) = 0$, by linearity and Adam's law.)

Proof

Prediction Perspective

- Predict or estimate the future observations or unknown parameters based on data
- $E(Y|X)$ is our **best predictor** of Y based on X.
- Best means it is the function of X with the lowest **mean squared error** (expected squared difference between Y and prediction of Y).

Proof Sketch

Orthogonality Property of MMSE

Theorem

(a) For any function $\phi(\cdot)$, one has

$$E[(Y - E[Y|X])\phi(X)] = 0$$

(b) Moreover, if the function $g(X)$ is such that

$$E[(Y - g(X))\phi(X)] = 0, \forall \phi(\cdot).$$

then $g(X) = E(Y|X)$

MMSE for Jointly Gaussian(Normal) Random Variables

Theorem

Let X, Y be jointly Gaussian random variables. Then

$$E[Y|X] = L[Y|X] = E(Y) + \frac{Cov(X,Y)}{Var(X)}(X - E(X)).$$

Remark: Statistical Learning Perspective

- In general, MMSE is a highly nonlinear function.
- Adoption of various approximation methods leads to various learning methods
  - Linear regression
  - Logistic regression
  - Polynomial regression
  - Regression with Spline functions
  - Neural network

Conditional Expectation as Projection Geometry

1 Hilbert space setting

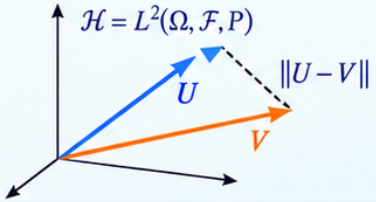
Hilbert space: $\mathcal{H} = L^2(\Omega, \mathcal{F}, \mathbb{P})$

Inner product: $\langle U, V \rangle = \mathbb{E}[UV]$

Orthogonal: $U \perp V$ iff $\langle U, V \rangle = \mathbb{E}[UV] = 0$.

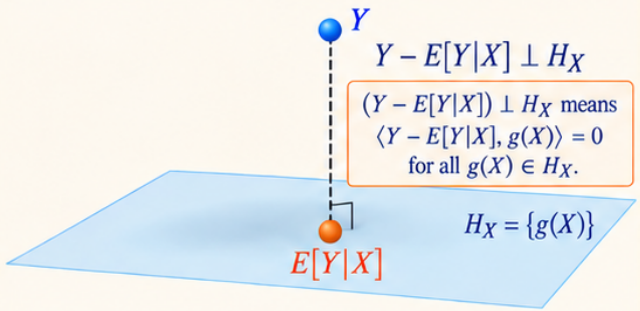
Squared geometric distance = mean squared error:

$$\|U - V\|^2 = E\left[(U - V)^2\right]$$



Ordinary expectation = projection onto constants.

2 Orthogonal projection

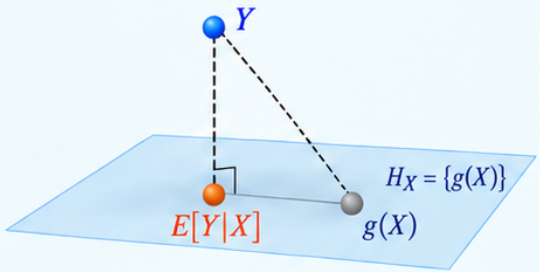


Conditional expectation = orthogonal projection

$$E[Y|X] = P_{H_X} Y$$

$$Y = E[Y|X] + \left(Y - E[Y|X]\right)$$

3 Optimal prediction by Pythagorean theorem

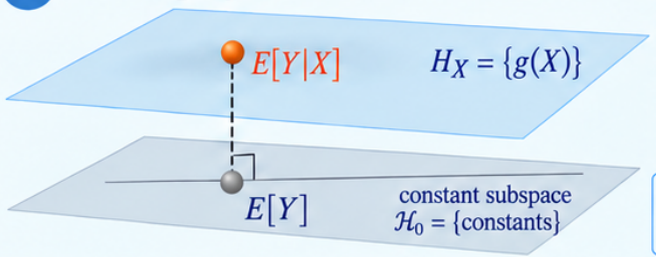


$$E[Y|X] = \arg\min_{g(X)} E\left[(Y - g(X))^2\right]$$

$$E\left[(Y - g(X))^2\right] = E\left[(Y - E[Y|X])^2\right] + E\left[(E[Y|X] - g(X))^2\right]$$

Projection gives the best predictor from X.

4 Nested projections and the law of iterated expectation



Iterated expectation = nested projections

Project onto H_X first, then onto constants.
Projecting onto a larger subspace first does not change the later projection onto a smaller subspace inside it.

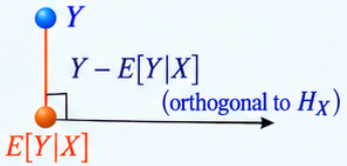
$$E(E[Y|X]) = E[Y]$$

$$E[Y] = P_{\text{constants}} Y = P_{H_0} Y, \quad \mathcal{H}_0 = \{\text{constants}\}$$

Expectation = projection onto constants.

Total variance = Pythagorean theorem

$$\text{Var}(Y) = \text{Var}(E[Y|X]) + E[\text{Var}(Y|X)]$$



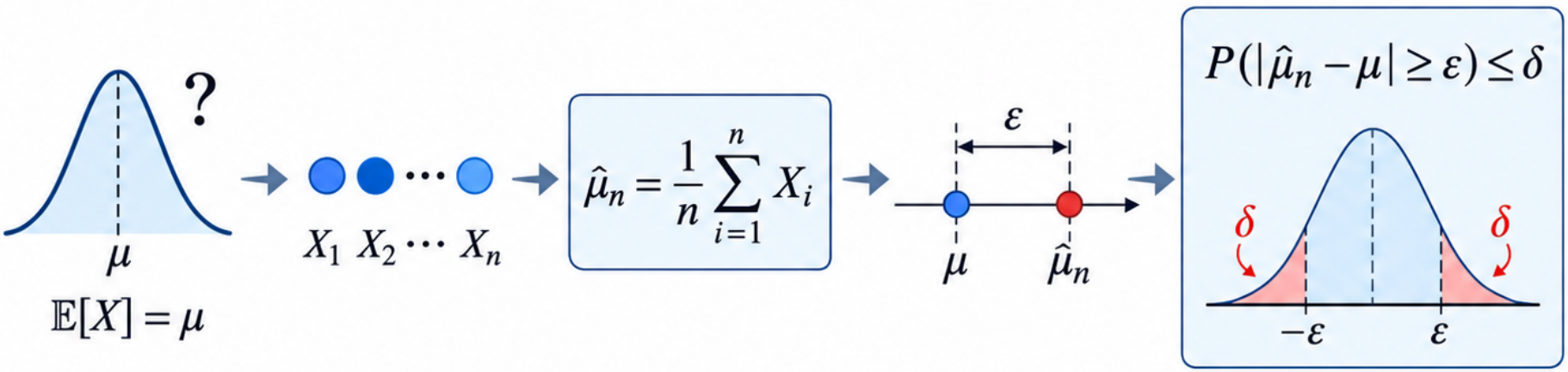
Summary

Conditional expectation = orthogonal projection onto H_X.
Ordinary expectation = projection onto the constant subspace.
Iterated expectation = nested projections.
Total variance = Pythagorean theorem.

Outline

1. Review of Probability & Statistics
2. Expectation & Conditional Expectation
3. Conditional Expectation as Projection Geometry
4. Concentration Inequality
5. Markov Dynamics I
6. Markov Dynamics II
7. Statistical Inference
8. Bayesian Learning: RL Perspective
9. Summary

Motivation


?
μ
$\mathbb{E}[X] = \mu$
X_1 X_2 $\cdots$ X_n
$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$
ε
μ
$\hat{\mu}_n$
$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \delta$
δ
δ
$-\varepsilon$
ε

Monte Carlo Estimation

Suppose $X_1, \ldots, X_n$ are i.i.d. samples from the same distribution as X.

Monte Carlo estimator (Sample Mean)

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Unbiased

$$\mathbb{E}[\hat{\mu}_n] = \mu.$$

Variance shrinks

$$\mathrm{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}.$$

Core idea

Monte Carlo replaces an expectation by an empirical average.

Monte Carlo Value Estimation in RL

Goal

Estimate

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s].$$

From sampled episodes

If we see $N(s)$ visits to state s and returns $G^{(1)}, \ldots, G^{(N(s))}$,

$$\hat{V}(s) = \frac{1}{N(s)} \sum_{i=1}^{N(s)} G^{(i)}.$$

Limitation

MC waits for a full return.

Small Example: Estimate a Value

A state s is visited in five episodes. The observed returns are

$$4, \quad 2, \quad 5, \quad -1, \quad 0.$$

Monte Carlo estimate

$$\hat{V}(s) = \frac{4 + 2 + 5 - 1 + 0}{5} = 2.$$

Small Example: Estimate a Value

A state s is visited in five episodes. The observed returns are

$$4, \quad 2, \quad 5, \quad -1, \quad 0.$$

Monte Carlo estimate

$$\hat{V}(s) = \frac{4 + 2 + 5 - 1 + 0}{5} = 2.$$

Question

If the true variance of returns is large, should we trust $\hat{V}(s) = 2$ after only five visits?

Law of Large Numbers

Statement

For i.i.d. samples $X_1, X_2, \ldots$ with mean μ,

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \to \mu$$

with high probability / almost surely under standard assumptions.

What it tells us

More samples eventually help.

What it does not tell us

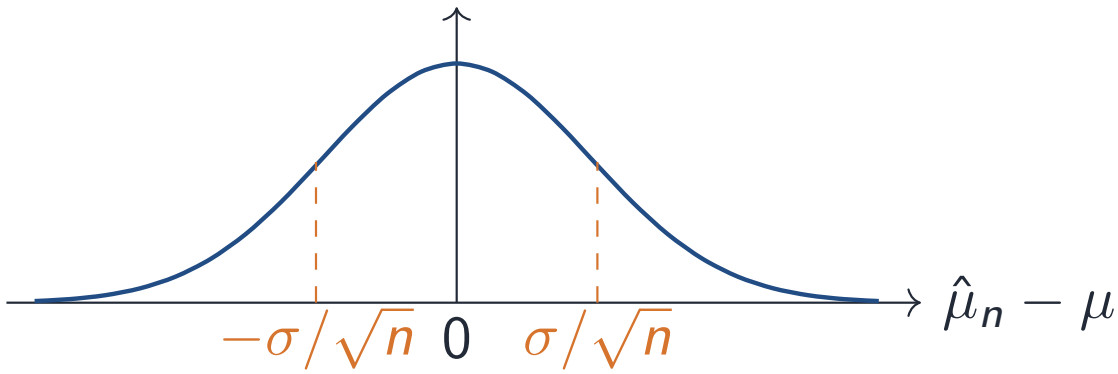
It does not directly say how many samples are enough at finite n.

Central Limit Theorem: Shape of the Error

Approximate distribution

For large n,

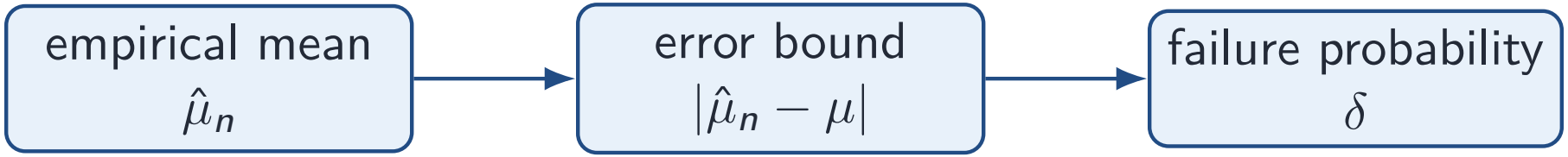
$$\frac{\hat{\mu}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0,1).$$



Core idea

The typical error scale is $O(1/\sqrt{n})$.

Why Concentration?



Concentration answers

$$\mathbb{P}(|\hat{\mu}_n - \mu| \geq \epsilon) \leq ?$$

RL use

Optimism, exploration bonuses, safe policy evaluation, and regret analysis all depend on finite-sample confidence.

Cauchy-Schwarz Inequality

Theorem

For any r.v.s X and Y with finite variances,

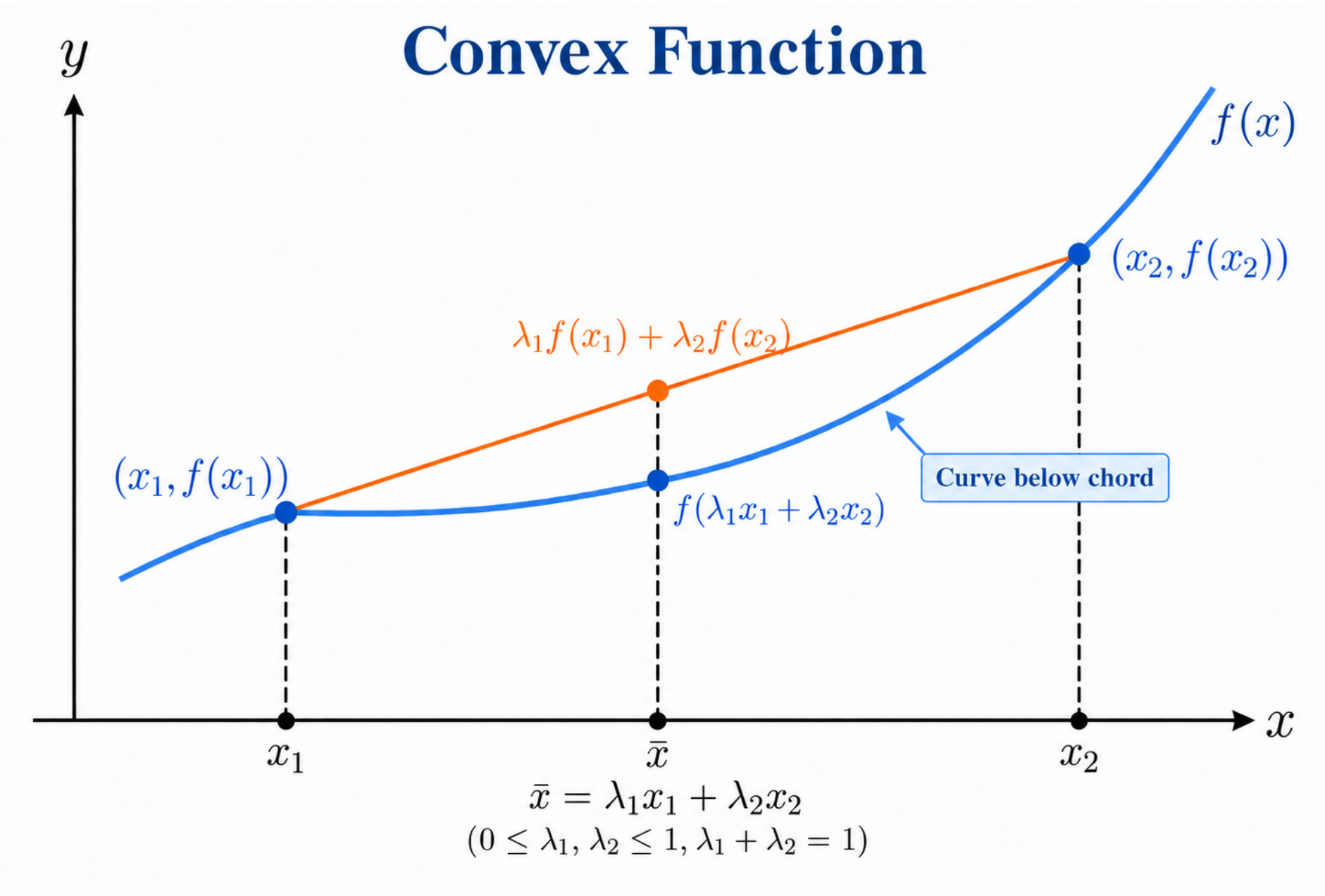
$$|E(XY)| \leq \sqrt{E(X^2)\,E(Y^2)}.$$

Jensen's Inequality

If f is a convex function, $0 \leq \lambda_1, \lambda_2 \leq 1, \lambda_1 + \lambda_2 = 1$, then for any x_1, x_2,

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2).$$

Jensen's Inequality



Jensen's Inequality

Theorem

Let X be a random variable. If g is a convex function, then $E(g(X)) \geq g(E(X))$. If g is a concave function, then $E(g(X)) \leq g(E(X))$. In both cases, the only way that equality can hold is if there are constants a and b such that $g(X) = a + bX$ with probability 1.

Quick Examples

Entropy

- Let X be a discrete r.v. whose distinct possible values are $a_1, a_2, ..., a_n$, with probabilities $p_1, p_2..., p_n$ respectively (so $p_1 + p_2 + \cdots + p_n = 1$).
- The *entropy* of X is defined as follows: $H(X) = \sum_{j=1}^{n} p_j \log_2(1/p_j)$.
- Using Jensen's inequality, show that the maximum possible entropy for X is when its distribution is uniform over $a_1, a_2, \ldots, a_n$, i.e., $p_j = 1/n$ for all j.
- This makes sense intuitively, since learning the value of X conveys the most information on average when X is equally likely to take any of its values, and the least possible information if X is a constant.

Proof

Kullback-Leibler Divergence

Let $\mathbf{p} = (p_1, ..., p_n)$ and $\mathbf{r} = (r_1, ..., r_n)$ be two probability vectors (so each is nonnegative and sums to 1). Think of each as a possible PMF for a random variable whose support consists of n distinct values. The *Kullback-Leibler* divergence between $\mathbf{p}$ and $\mathbf{r}$ is defined as

$$D(\mathbf{p}, \mathbf{r}) = \sum_{j=1}^{n} p_j \log_2(1/r_j) - \sum_{j=1}^{n} p_j \log_2(1/p_j).$$

Show that the Kullback-Leibler divergence is nonnegative.

Proof

Markov's Inequality

Theorem

For any r.v. X and constant $a > 0$,

$$P(|X| \geq a) \leq \frac{E|X|}{a}.$$

Proof

Chebyshev's Inequality

Theorem

Let X have mean μ and variance σ^2. Then for any $a > 0$,

$$P\left(|X - \mu| \geq a\right) \leq \frac{\sigma^2}{a^2}.$$

Chernoff's Inequality

Theorem

For any r.v. X and constants $a > 0$ and $t > 0$,

$$P(X \geq a) \leq \frac{E\left(e^{tX}\right)}{e^{ta}}.$$

Chernoff's Technique

Theorem

For any r.v. X and constants a,

$$P(X \geq a) \leq \inf_{t>0} \frac{E\left(e^{tX}\right)}{e^{ta}}$$

$$P(X \leq a) \leq \inf_{t<0} \frac{E\left(e^{tX}\right)}{e^{ta}}.$$

Hoeffding Lemma

Lemma

Let the random variable X satisfy $\mathbb{E}(X) = 0$ and $a \leq X \leq b$, where a and b are constants. Then for any $\lambda > 0$,

$$\mathbb{E}(e^{\lambda X}) \leq e^{\frac{1}{8}\lambda^2(b-a)^2}.$$

Useful Analysis Tools

- Jensen's inequality: if f is convex, $0 \leq \lambda_1, \lambda_2 \leq 1, \lambda_1 + \lambda_2 = 1$, then for any x_1, x_2,

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2).$$

- Taylor's theorem or Taylor's expansion: If all the derivatives of a function $f(x)$ exist at point a, then for any positive integer k, there exist a real number θ between a and x such that

$$f(x) = f(a) + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(k+1)}(\theta)}{(k+1)!}(x-a)^{k+1}.$$

Hoeffding Inequality

Theorem

Let the random variables $X_1, X_2, \ldots, X_n$ be independent with $E(X_i) = \mu$, $a \leq X_i \leq b$ for each $i = 1, \ldots, n$, where a, b are constants. Then for any $\epsilon \geq 0$,

$$\mathbb{P}(|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu| \geq \epsilon) \leq 2e^{-\frac{2n\epsilon^2}{(b-a)^2}}.$$

Proof Sketch

More General Hoeffding Inequality

Theorem

Let the random variables $X_1, X_2, \ldots, X_n$ be independent, with $a_k \leq X_k \leq b_k$ for each k, where a_k, b_k are constants. Let $S_n = \sum_{k=1}^{n} X_k$ and let $\mu = \mathbb{E}(S_n)$. Then for any $t \geq 0$,

$$\mathbb{P}(|S_n - \mu| \geq t) \leq 2e^{-\frac{2t^2}{\sum_{k=1}^{n}(b_k - a_k)^2}}.$$

Hoeffding Inequality

Bounded independent samples

If $X_i \in [a, b]$ are independent and $\mathbb{E}[X_i] = \mu$, then

$$\boxed{\mathbb{P}(|\hat{\mu}_n - \mu| \geq \epsilon) \leq 2 \exp\left(-\frac{2n\epsilon^2}{(b-a)^2}\right)}$$

Reward in $[0, 1]$

$$\mathbb{P}(|\hat{\mu}_n - \mu| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

From Hoeffding to Confidence Radius

For rewards in $[0, 1]$:

$$\mathbb{P}(|\hat{\mu}_n - \mu| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

Set the failure probability to δ:

$$2e^{-2n\epsilon^2} = \delta.$$

Solve for ϵ:

$$\epsilon = \sqrt{\frac{\log(2/\delta)}{2n}}.$$

Confidence interval

$$\boxed{\mu \in \left[\hat{\mu}_n - \sqrt{\frac{\log(2/\delta)}{2n}}, \hat{\mu}_n + \sqrt{\frac{\log(2/\delta)}{2n}}\right]}$$

with probability at least $1 - \delta$.

Picture: Estimate plus Uncertainty

Core idea

Small n gives wide confidence intervals; large n gives narrow ones.

Small Example: How Many Samples?

For rewards in $[0, 1]$, Hoeffding gives

$$\mathbb{P}(|\hat{\mu}_n - \mu| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

Question

How large should n be so that error is at most 0.1 with confidence 95%?

Small Example: How Many Samples?

For rewards in $[0, 1]$, Hoeffding gives

$$\mathbb{P}(|\hat{\mu}_n - \mu| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

Question

How large should n be so that error is at most 0.1 with confidence 95%?

Set $\epsilon = 0.1$ and $\delta = 0.05$:

$$2e^{-2n(0.1)^2} \leq 0.05.$$

$$n \geq \frac{\log(40)}{2(0.1)^2} \approx 184.4.$$

$$\boxed{n \geq 185}$$

Preview: Confidence Becomes Exploration

Empirical estimate

$$\hat{\mu}_a$$

Confidence bonus

$$\sqrt{\frac{2 \log t}{N_t(a)}}$$

Optimistic action selection: upper confidence bound (UCB)

$$A_t = \arg\max_a \left[\hat{\mu}_a + \sqrt{\frac{2 \log t}{N_t(a)}} \right]$$

This is Week 5

UCB is concentration turned into an exploration algorithm.

Summary

From expectation to finite-sample trust

$$\mu = \mathbb{E}[X] \qquad \text{unknown population quantity}$$

$$\widehat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \text{observable sample estimate}$$

$$\boxed{\Pr\left(|\widehat{\mu}_n - \mu| \geq \varepsilon\right) \leq \delta}$$

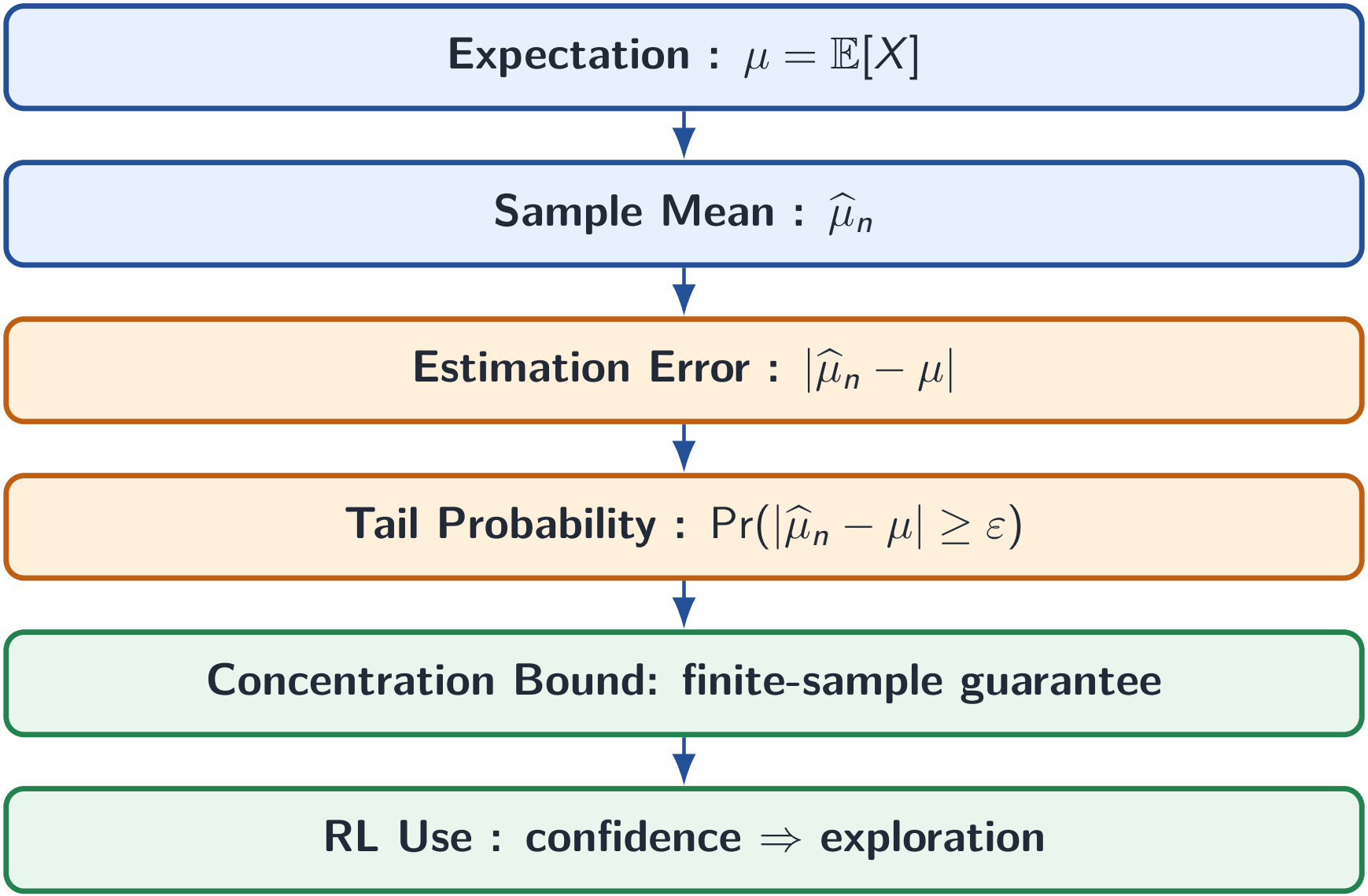
Key motivation

LLN says: eventually correct.

Concentration says: reliable with finite samples.

Concentration = finite samples + high probability + confidence.

Summary


Expectation : $\mu = \mathbb{E}[X]$
Sample Mean : $\widehat{\mu}_n$
Estimation Error : $|\widehat{\mu}_n - \mu|$
Tail Probability : $\Pr(|\widehat{\mu}_n - \mu| \geq \varepsilon)$
Concentration Bound: finite-sample guarantee
RL Use : confidence $\Rightarrow$ exploration

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Stochastic Process

Definition

A stochastic process is a sequence of random variables:

$$X_0, X_1, X_2, \ldots$$

indexed by time.

RL examples

- state process: $S_0, S_1, S_2, \ldots$
- action process: $A_0, A_1, A_2, \ldots$
- reward process: $R_1, R_2, R_3, \ldots$

Core idea

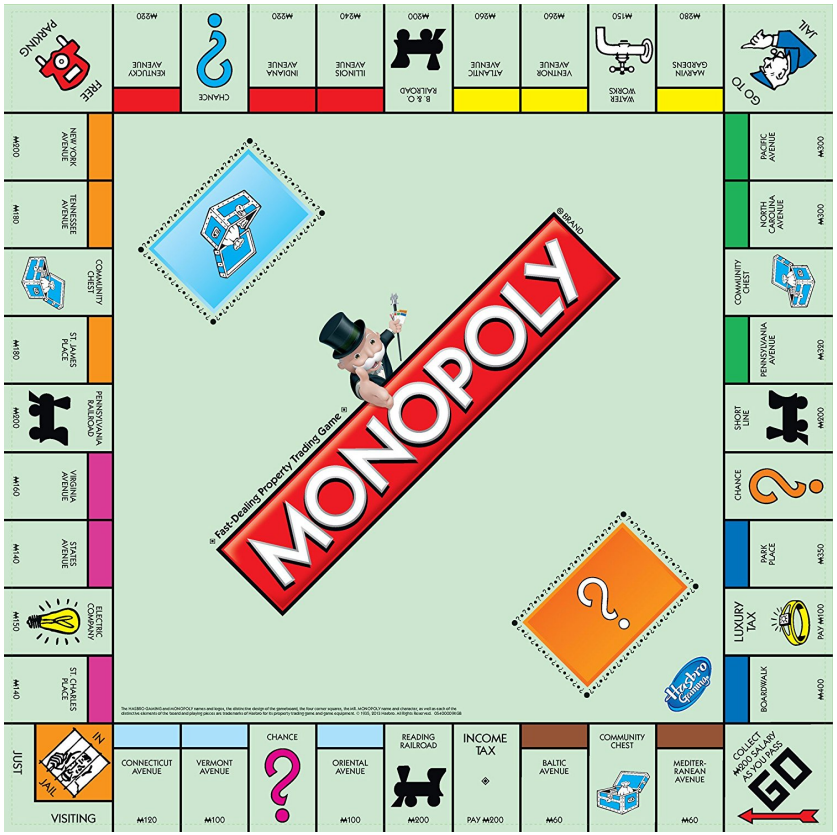
Sequential learning is learning from a stochastic process, not from a static dataset.

More General Definition

- A stochastic process is a collection of random variables $\{X_t, t \in I\}$. The set I is the *index set* of the process. The random variables are defined on a common state space $\mathcal{S}$.
- I is discrete: discrete-time stochastic processes (sequences of random variables)
- I is continuous: continuous-time stochastic processes (uncountable collections of random variables)

Example: Discrete Time & Discrete State Space

- State Space:$\{1, \ldots, 40\}$
- X_k: the player's board position after k dice rollings.
- Stochastic Process for Monopoly: $X_0, X_1, \ldots$



Example: Discrete Time & Continuous State Space

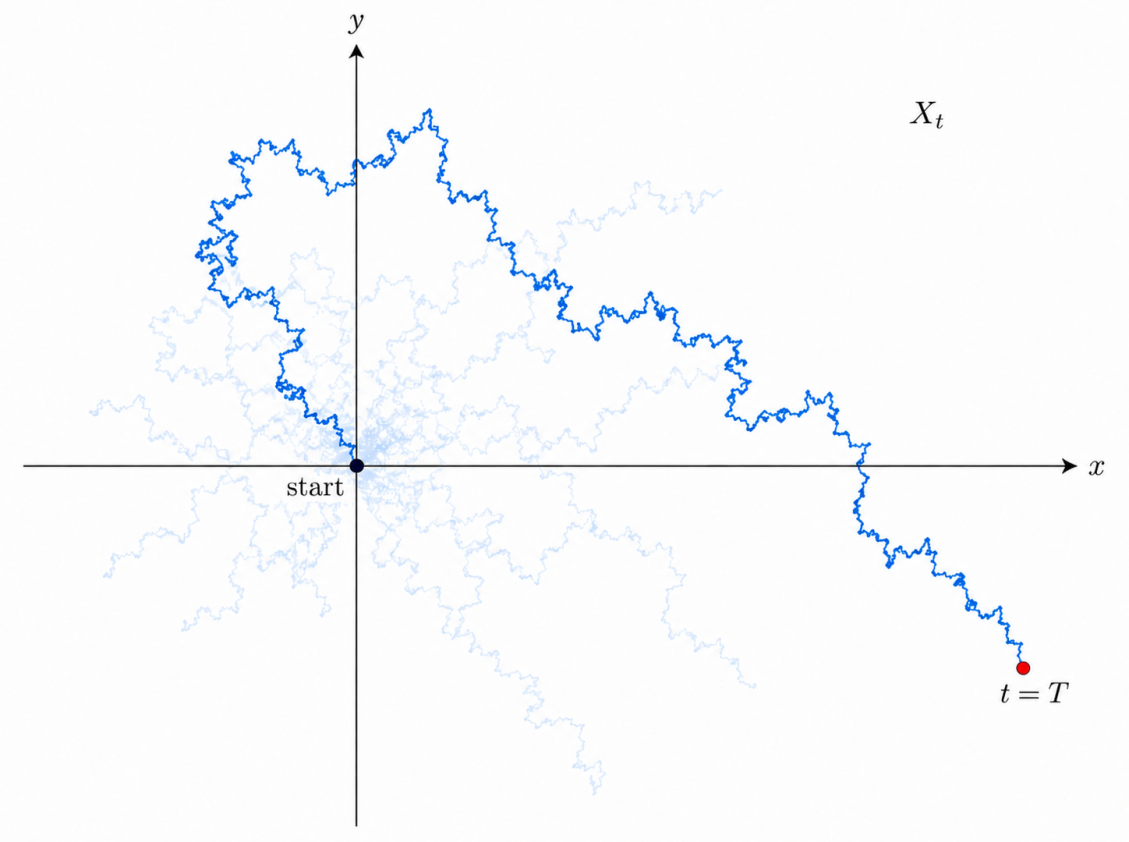
- Air-monitoring with PM2.5 measurements every hour
- State Space:$(0, 2000)$
- X_k: the PM2.5 measurement at the kth hour.
- Stochastic Process for Air-monitoring: $X_0, X_1, \ldots$

Example: Continuous Time & Discrete State Space

- We receive emails at random times day and night.
- State Space:$\{0, 1, 2, \ldots\}$
- $X_t, t \in [0, \infty)$: the number of emails we receive up to time t
- Stochastic Process for Email: $\{X_t\}$

Example: Continuous Time & Continuous State Space

- Two-dimensional Brownian Motion
- State Space: $\mathbb{R}^2$
- $X_t, t \in [0, \infty)$: position of the particle at time t
- Stochastic Process for random motion of particles: $\{X_t\}$



Model Selection in Stochastic Modeling

- Enough complexity to capture the complexity of the phenomena in question
- Enough structure and simplicity to allow one to compute things of interest

Motivation

- Introduced by Andrey Markov in 1906
- IID sequence of random variables: too restrictive assumption
- Completely dependent among random variables: hard to analysis
- Markov chain: happy medium between complete independence & complete dependence.

Markov Model

Three basic components of Markov model

- A sequence of random variables $\{X_t, t \in \mathcal{T}\}$, where $\mathcal{T}$ is an index set, usually called "**time**".
- All possible sample values of $\{X_t, t \in \mathcal{T}\}$ are called "**states**", which are elements of a sate space $\mathcal{S}$.
- "**Markov property**": given the present value(information) of the process, the future evolution of the process is independent of the past evolution of the process.

Classification of Markov Model

- *Discrete-Time Markov Chain*: Discrete $\mathcal{S}$ & Discrete $\mathcal{T}$
- *Continuous-Time Markov Chain*: Discrete $\mathcal{S}$ & Continuous $\mathcal{T}$
- *Discrete Markov Process*: Continuous $\mathcal{S}$ & Discrete $\mathcal{T}$
- *Continuous Markov Process*: Continuous $\mathcal{S}$ & Continuous $\mathcal{T}$

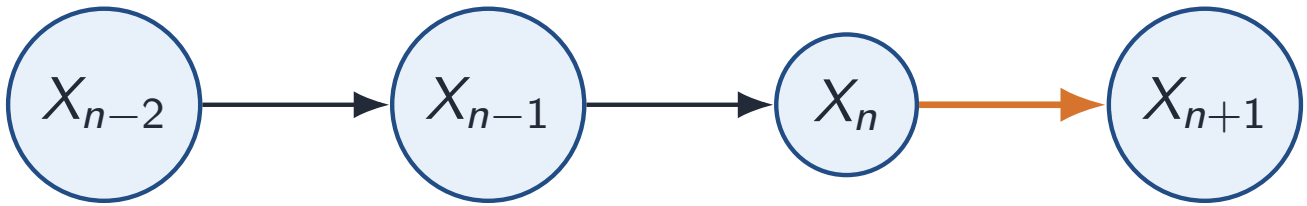
Our focus: Discrete(Continuous)-Time Markov Chain with finite state space

Markov Chain

Definition

A sequence of random variables $X_0, X_1, X_2, ...$ taking values in the *state space* $\{1, 2, ..., M\}$ is called a *Markov chain* if for all $n \geq 0$,

$$P\left(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0\right) = P\left(X_{n+1} = j | X_n = i\right).$$



Time-homogeneous Markov Chains

Definition

Given a Markov chain $X_0, X_1, X_2, \ldots$ It is called time-homogeneous Markov chain if for all $n \geq 0$,

$$P\left(X_{n+1} = j | X_n = i\right) = q_{i,j}.$$

where $q_{i,j}$ is a constant independent of n.

From now on, we focus on time-homogeneous Markov Chains, and we call it Markov chain in brevity.

Transition Matrix

Definition

Let $X_0, X_1, X_2, ...$ be a Markov chain with state space $\{1, 2, ..., M\}$, and let $q_{i,j} = P(X_{n+1} = j | X_n = i)$ be the transition probability from state i to state j. The $M \times M$ matrix $Q = (q_{i,j})$ is called the *transition matrix* of the chain.

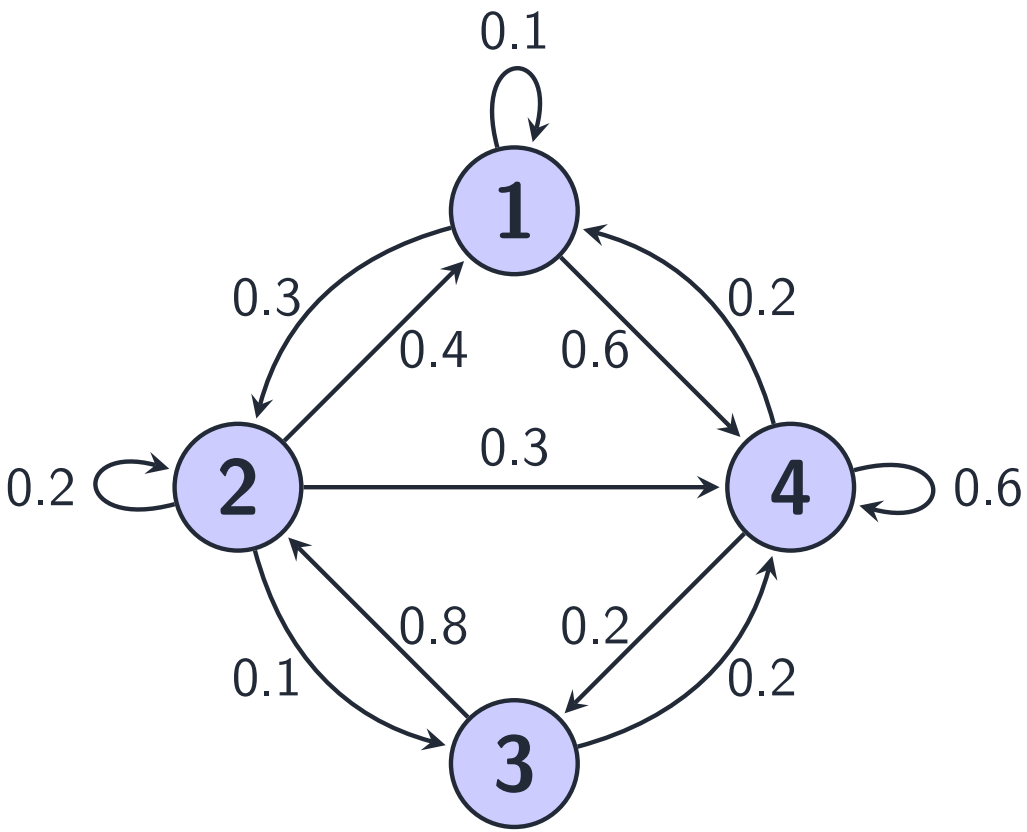
Each row is a probability distribution

$$q_{i,j} \geq 0, \qquad \sum_j q_{i,j} = 1.$$

Two-state chain

$$Q = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$$

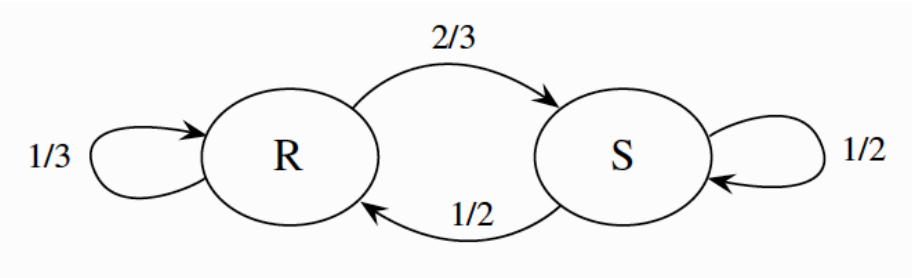
Graphical and Matrix Form of Markov Chain



$$Q = \begin{bmatrix} 0.1 & 0.3 & 0 & 0.6 \\ 0.4 & 0.2 & 0.1 & 0.3 \\ 0 & 0.8 & 0 & 0.2 \\ 0.2 & 0 & 0.2 & 0.6 \end{bmatrix}$$

Row sum $= 1$ in Transition Matrix.

Example: Rainy-Sunny Markov Chain

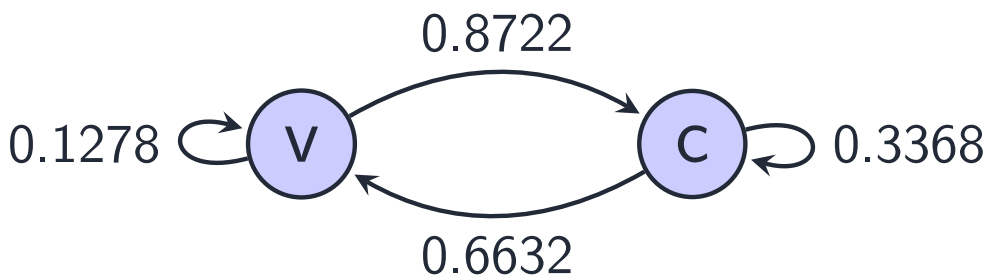


$$\begin{array}{c} \\ R \\ S \end{array} \begin{array}{c} \begin{array}{cc} R & S \end{array} \\ \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \end{array}$$

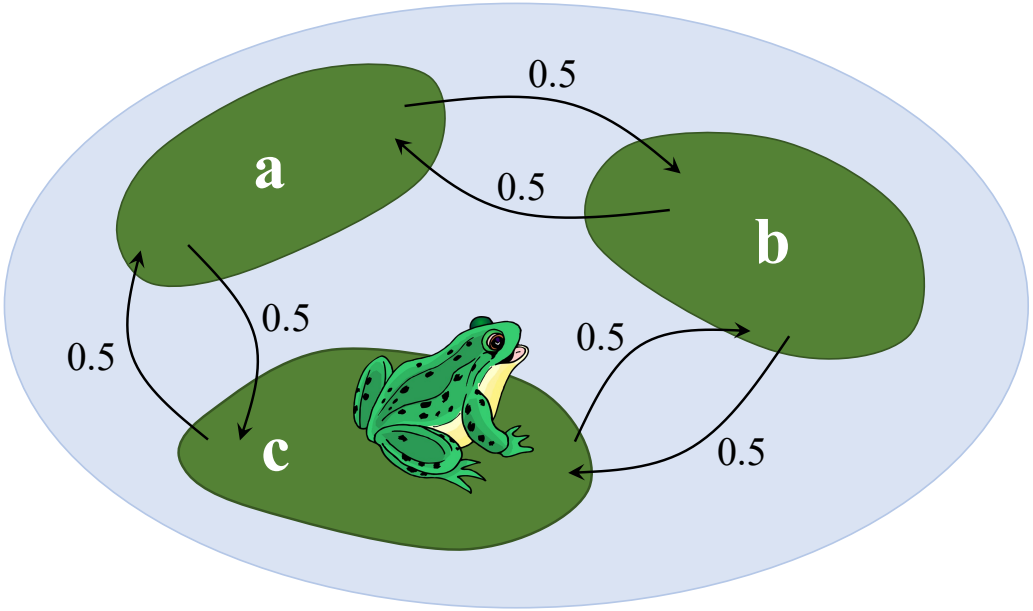
Example: The First Markov Chain in History

- Andrey Andreyevich Markov was interested in investigating the way the vowels and consonants alternate in Russian literature, e.g., "Eugene Onegin"by Pushkin
- He classified 20,000 consecutive characters: 8638 vowels & 11362 consonants

$$\begin{array}{c} \\ \text{vowel} \\ \text{consonant} \end{array} \begin{array}{c} \begin{array}{cc} \text{vowel} & \text{consonant} \end{array} \\ \begin{bmatrix} 1104/8638 & 7534/8638 \\ 7535/11362 & 3827/11362 \end{bmatrix} \end{array} = \begin{bmatrix} 0.1278 & 0.8722 \\ 0.6632 & 0.3368 \end{bmatrix}$$



Example: Random Walk on A Graph



$$\begin{array}{c} \\ a \\ b \\ c \end{array} \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix} \end{array}$$

n-step Transition Probability

Definition

Let $X_0, X_1, X_2, \ldots$ be a Markov chain with transition matrix Q. The n-step transition probability from i to j is the probability of being at j exactly n steps after being at i. We denote this by $q_{i,j}^{(n)}$:

$$q_{i,j}^{(n)} = P\left(X_n = j | X_0 = i\right).$$

Example: 2-step Transition Probability

$$q_{i,j}^{(2)} = P\left(X_2 = j | X_0 = i\right) = \sum_k q_{i,k} q_{k,j} = \text{(i,j) entry of } Q^2.$$

Chapman-Kolmogorov Relationship

$$q_{i,j}^{(m+n)} = P\left(X_{m+n} = j | X_0 = i\right) = \sum_k q_{i,k}^{(m)} q_{k,j}^{(n)} = \text{(i,j) entry of } Q^{m+n}.$$

Distribution Evolution

Let μ_n be the row vector distribution of X_n:

$$\mu_n(i) = \mathbb{P}(X_n = i).$$

Then

$$\boxed{\mu_{n+1} = \mu_n Q.}$$

By iteration,

$$\boxed{\mu_n = \mu_0 Q^n.}$$

RL connection

A policy plus environment dynamics induces a Markov chain over states.

Small Example: Two-State Weather Chain

States: Sunny (S) and Rainy (R).

$$Q = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}.$$

Start sunny:

$$\mu_0 = (1, 0).$$

After one day:

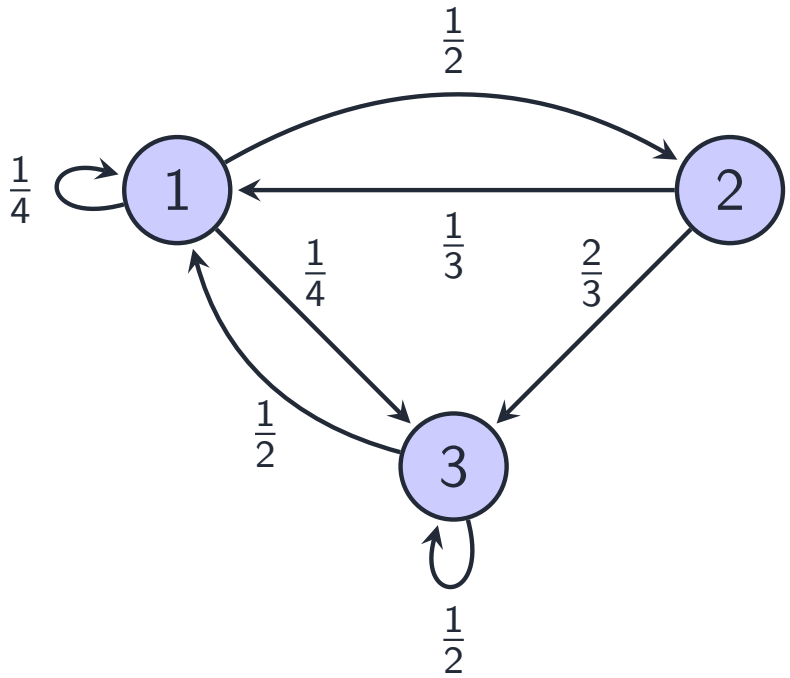
$$\mu_1 = \mu_0 Q = (0.8, 0.2).$$

After two days:

$$\mu_2 = \mu_1 Q = (0.8, 0.2)Q = (0.70, 0.30).$$

Example

Given a Markov chain $X_0, X_1, X_2, \ldots$ with state space $\mathcal{S} = \{1, 2, 3\}$.

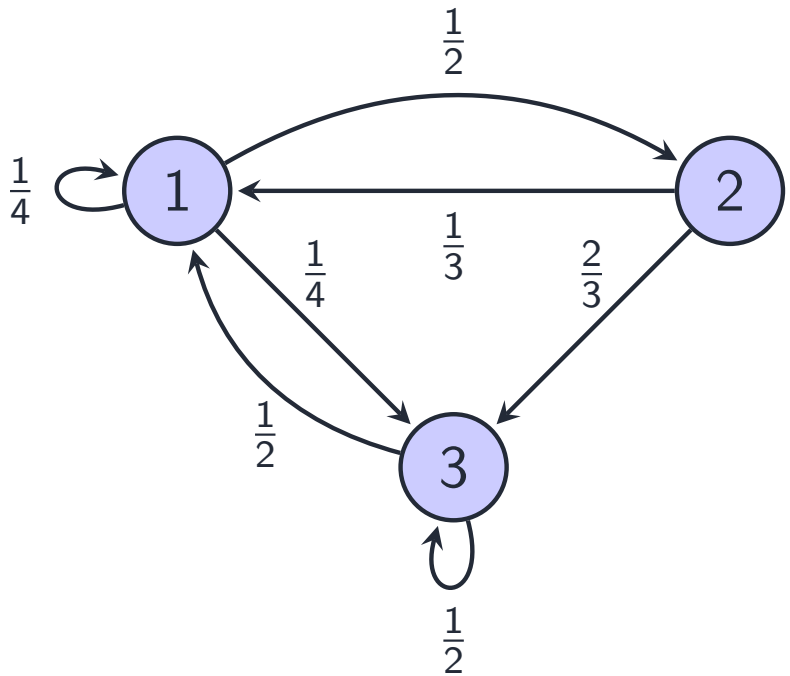


$$
\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}
\begin{array}{c}
\begin{array}{ccc} 1 & 2 & 3 \end{array} \\
\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}
\end{array}
$$

- Find $P(X_3 = 1 | X_2 = 1)$ and $P(X_4 = 3 | X_3 = 2)$.

Example

Given a Markov chain $X_0, X_1, X_2, \ldots$ with state space $\mathcal{S} = \{1, 2, 3\}$.

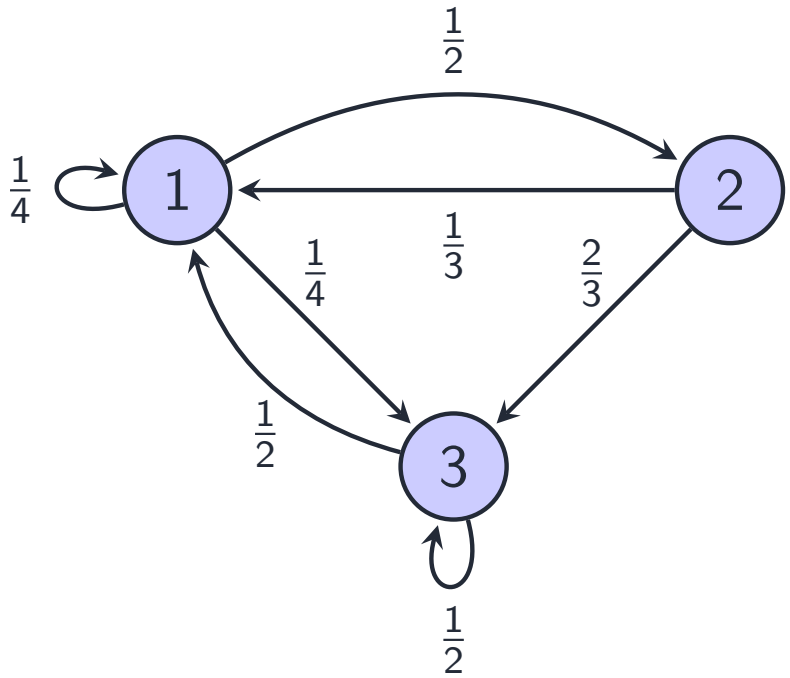


$$
\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}
\begin{array}{c}
\begin{array}{ccc} 1 & 2 & 3 \end{array} \\
\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}
\end{array}
$$

- If $P(X_0 = 1) = \frac{1}{3}$, find $P(X_0 = 1, X_1 = 2, X_2 = 3)$.

Example

Given a Markov chain $X_0, X_1, X_2, \ldots$ with state space $\mathcal{S} = \{1, 2, 3\}$.

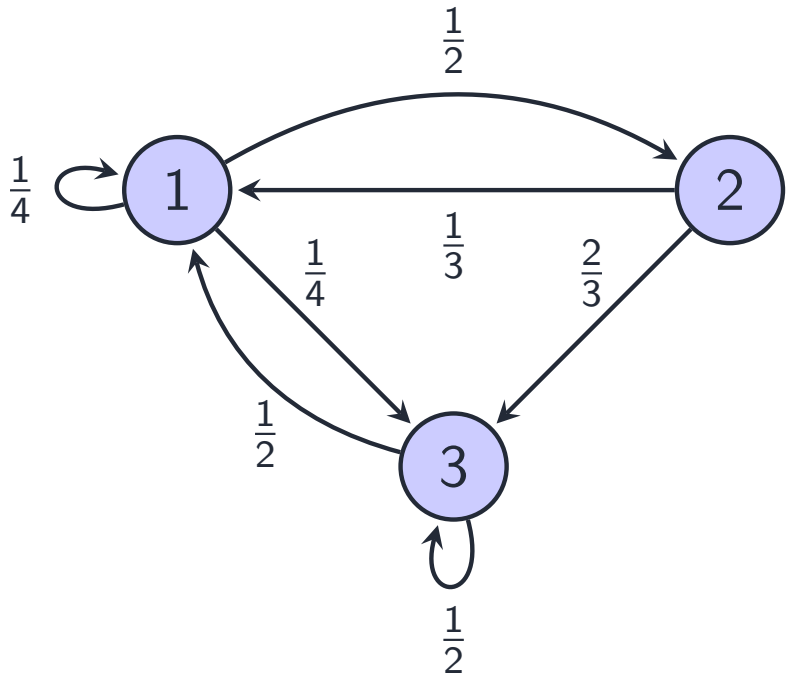


$$
\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}
\begin{array}{c}
\begin{array}{ccc} 1 & 2 & 3 \end{array} \\
\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}
\end{array}
$$

- Find $P(X_2 = 1 | X_0 = 1)$, $P(X_2 = 2 | X_0 = 1)$, and $P(X_2 = 3 | X_0 = 1)$.

Example

Given a Markov chain $X_0, X_1, X_2, \ldots$ with state space $\mathcal{S} = \{1, 2, 3\}$.



$$
\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}
\begin{array}{c}
\begin{array}{ccc} 1 & 2 & 3 \end{array} \\
\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}
\end{array}
$$

- Find $E(X_2|X_0 = 1)$.

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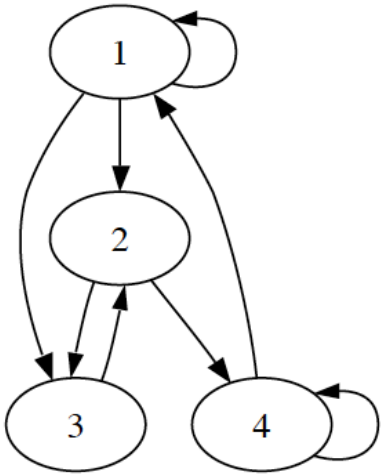
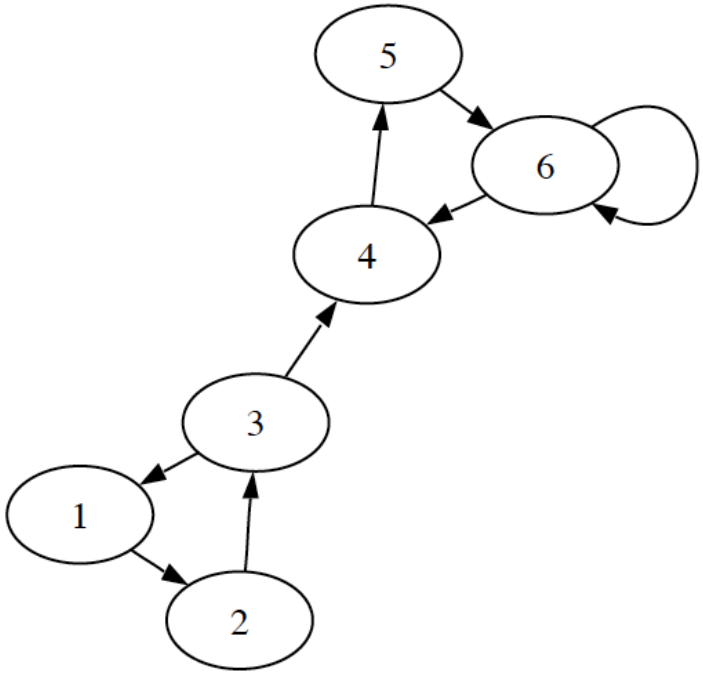
Recurrent and Transient States

Definition

State i of a Markov chain is **recurrent** if starting from i, the probability is 1 that the chain will eventually return to i. Otherwise, the state is **transient**, which means that if the chain starts from i, there is a positive probability of never returning to i.

Example





Irreducible and Reducible Chain

Definition

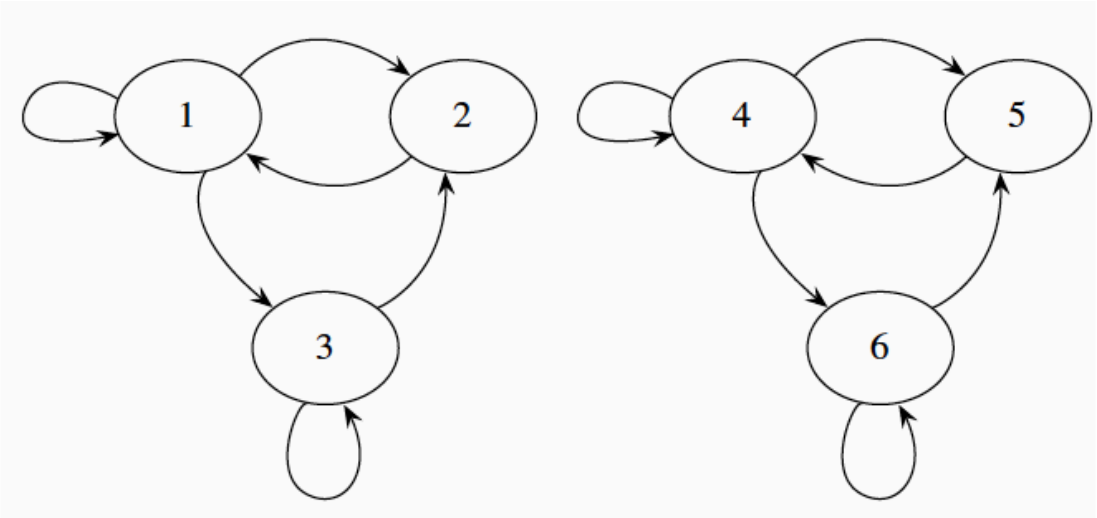
A Markov chain with transition matrix Q is *irreducible* if for any two states i and j, it is possible to go from i to j in a finite number of steps (with positive probability). That is, for any states i,j there is some positive integer n such that the (i, j) entry of Q^n is positive. A Markov chain that is not irreducible is called *reducible*.

Irreducible Implies All States Recurrent

Theorem

In an irreducible Markov chain with a finite state space, all states are recurrent.

A Reducible Markov Chain with Recurrent States



Period

Definition

For a Markov chain with transition matrix Q, the *period* of state i, denoted *d(i)*, is the greatest common divisor of the set of possible return times to i. That is,

$$d(i) = gcd\{n > 0 : Q^n_{i,i} > 0\}.$$

If $d(i) = 1$, state i is said to be *aperiodic*. If the set of return times is empty, set $d(i) = +\infty$.

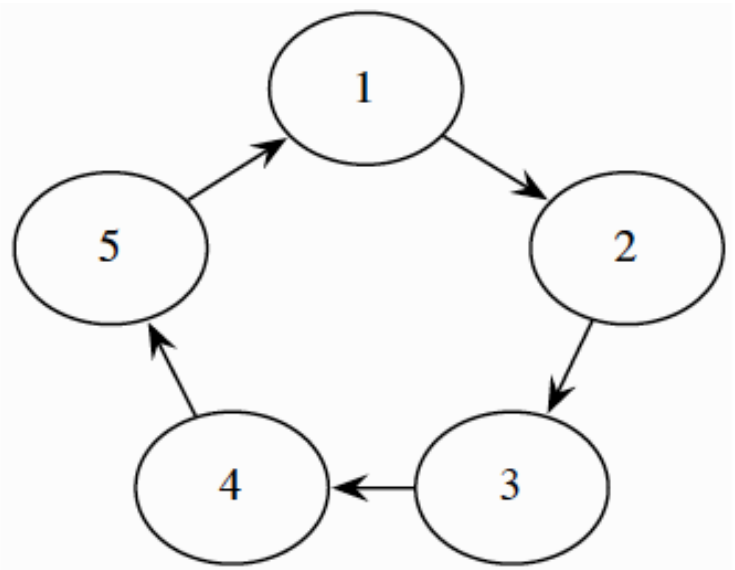
Periodic, Aperiodic Markov Chain

Definition

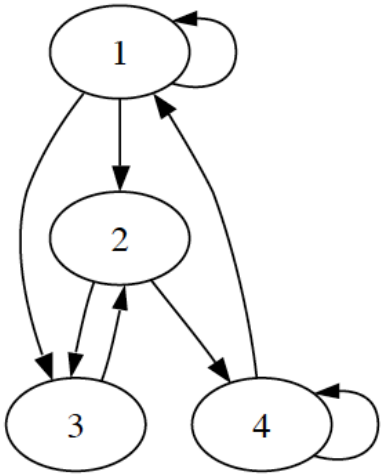
A Markov chain is called *periodic* if it is irreducible and all states have period greater than 1.

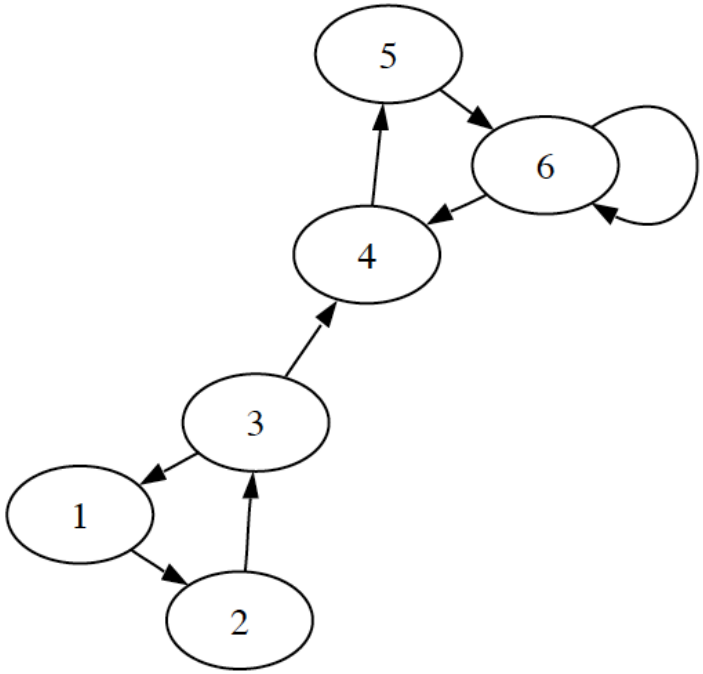
A Markov chain is called *aperiodic* if it is irreducible and all states have period equal to 1.

Example: Periodic Chain



Example





Stationary Distribution

Definition

A distribution π is stationary if

$$\boxed{\pi = \pi Q.}$$

Meaning

If $X_n \sim \pi$, then $X_{n+1} \sim \pi$.

RL preview

Under a fixed policy, the long-run state visitation pattern is often described by a stationary distribution or a discounted occupancy measure.

Compute the Stationary Distribution

For

$$Q = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}, \quad \pi = (p, 1-p).$$

Stationarity $\pi = \pi Q$ gives

$$p = 0.8p + 0.3(1-p).$$

Solve:

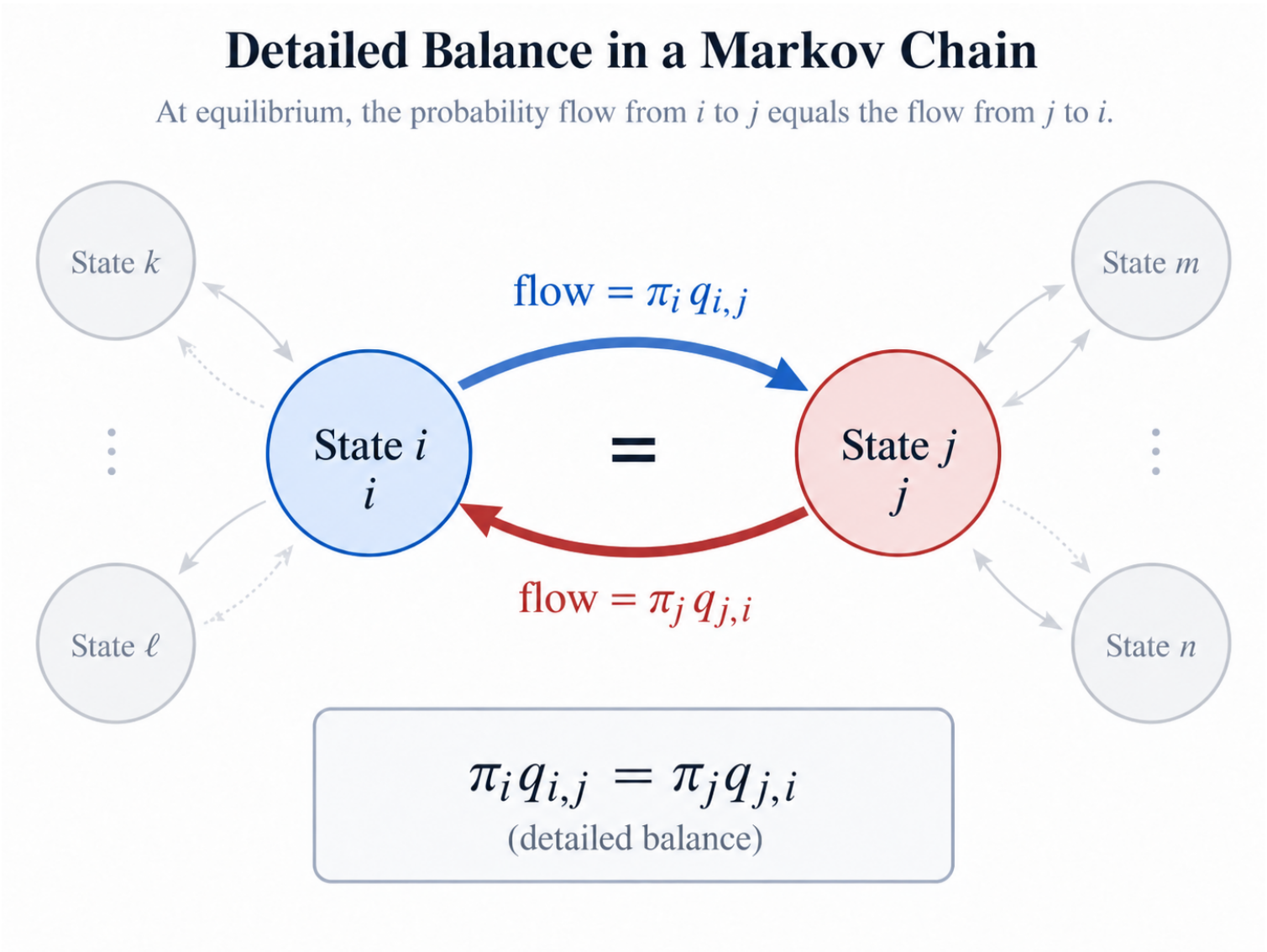
$$p = 0.8p + 0.3 - 0.3p = 0.5p + 0.3,$$

$$0.5p = 0.3, \qquad p = 0.6.$$

Thus

$$\boxed{\pi = (0.6, 0.4).}$$

Detailed Balance



Detailed Balance Implies Stationarity

For each target state i,

$$(\pi Q)_i = \sum_j \pi_j q_{j,i}.$$

By detailed balance,

$$\pi_j q_{j,i} = \pi_i q_{i,j}.$$

Therefore

$$(\pi Q)_i = \sum_j \pi_i q_{i,j} = \pi_i \sum_j q_{i,j} = \pi_i.$$

Conclusion

$$\boxed{\pi Q = \pi.}$$

Reversibility

Definition

Let $Q = (q_{i,j})$ be the transition matrix of a Markov chain. Suppose there is $\mathbf{s} = (s_1, ..., s_M)$ with $s_i \geq 0$, $\sum_i s_i = 1$, such that

$$s_i q_{i,j} = s_j q_{j,i}$$

for all states i and j. This equation is called the *reversibility* or *detailed balance* condition, and we say that the chain is *reversible* with respect to $\mathbf{s}$ if it holds.

Check the Detailed Balance Equation

Theorem

If for an irreducible Markov chain with transition matrix $Q = (q_{i,j})$, there exists a probability solution π to the detailed balance equations

$$\pi_i q_{i,j} = \pi_j q_{j,i}$$

for all pairs of states i, j, then this Markov chain is positive recurrent, time-reversible and the solution π is the unique stationary distribution.

Example: Symmetric Transition Matrix

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix} \end{array}$$

Theorem

If the transition matrix Q for an irreducible Markov chain is symmetric, then the uniform distribution over all states, $(1/M, 1/M, ..., 1/M)$, is the unique stationary distribution.

Mixing: Forgetting the Initial State

Suppose two chains start from different initial distributions:

$$\mu_0, \quad \nu_0.$$

After t steps:

$$\mu_t = \mu_0 Q^t, \qquad \nu_t = \nu_0 Q^t.$$

Mixing intuition

A chain mixes if

$$\mu_0 P^t \approx \nu_0 P^t \approx \pi$$

for large t.

RL implication

Slow mixing means experience remains highly correlated and exploration may be hard.

Total Variation Distance

Definition

For two distributions p and q on a finite space,

$$\|p - q\|_{\mathrm{TV}} = \frac{1}{2} \sum_x |p(x) - q(x)|.$$

Small example

$$p = (0.7, 0.3), \quad q = (0.6, 0.4).$$

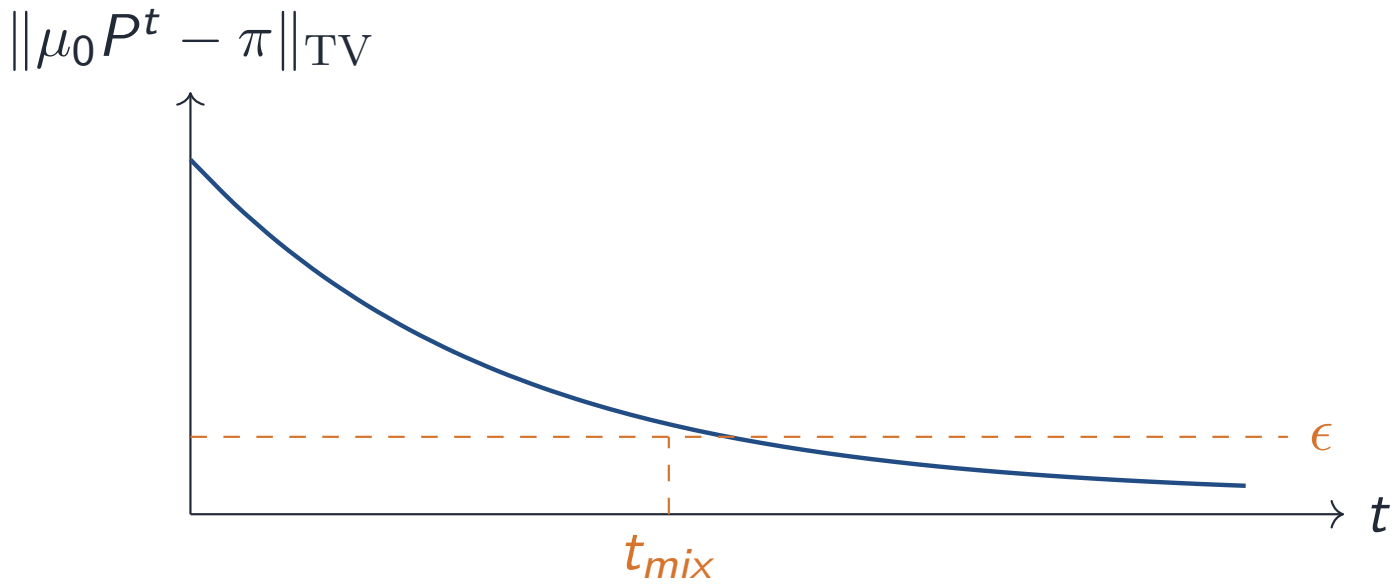
$$\|p - q\|_{\mathrm{TV}} = \frac{1}{2}(|0.1| + |-0.1|) = 0.1.$$

Mixing Time

Definition

A common definition is

$$t_{\text{mix}}(\epsilon) = \min\left\{ t : \max_{\mu_0} \|\mu_0 Q^t - \pi\|_{\text{TV}} \leq \epsilon \right\}.$$



Mixing Time: Fast vs. Slow

$$t_{\mathrm{mix}}(\epsilon) = \min\left\{ t : \max_{\mu_0} \|\mu_0 Q^t - \pi\|_{\mathrm{TV}} \le \epsilon \right\}$$

Fast mixing

$$t_{\mathrm{mix}}^{(n)}(\epsilon) = \mathrm{poly}\left(n, \log\frac{1}{\epsilon}\right)$$

- forgets the initial state
- no severe bottleneck

Slow mixing

$$t_{\mathrm{mix}}^{(n)}(\epsilon) = e^{\Omega(n)} \quad \text{(typical)}$$

- long memory
- bottleneck or metastability

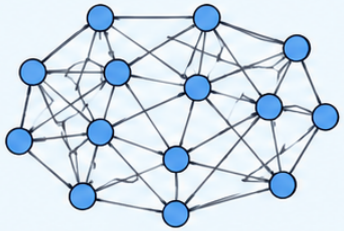
large conductance → **large spectral gap** → **fast mixing**

Takeaway: mixing time measures how fast the chain becomes insensitive to the starting distribution.

Fast vs Slow Mixing

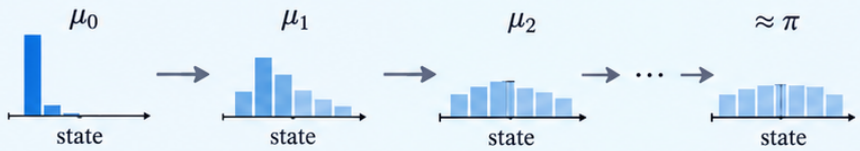
Fast Mixing

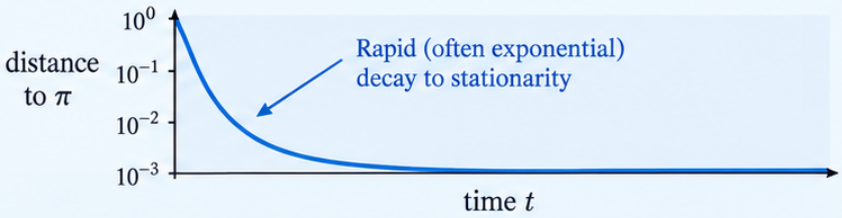
Well-connected state space, no bottleneck.



- small mixing time
- quickly forgets initial state
- large spectral gap
- good conductance

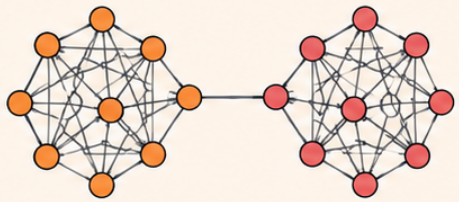
Distributions converge quickly





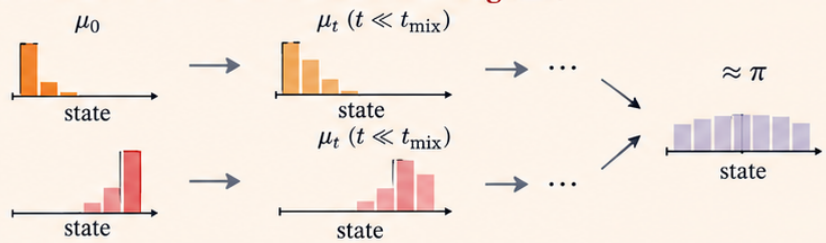
Slow Mixing

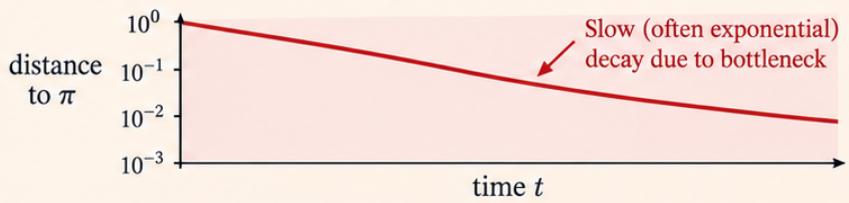
Poorly connected state space with a bottleneck.



- large mixing time
- long memory of initial state
- small spectral gap
- bottleneck / low conductance

Distributions remain different for a long time





Mixing time (total variation):

$$t_{\text{mix}}(\epsilon) = \min\left\{t : \max_{\mu_0} \|\mu_0 Q^t - \pi\|_{\text{TV}} \leq \epsilon\right\}$$

Fast mixing: polynomial growth in system size. **Slow mixing:** often exponential growth due to bottlenecks.

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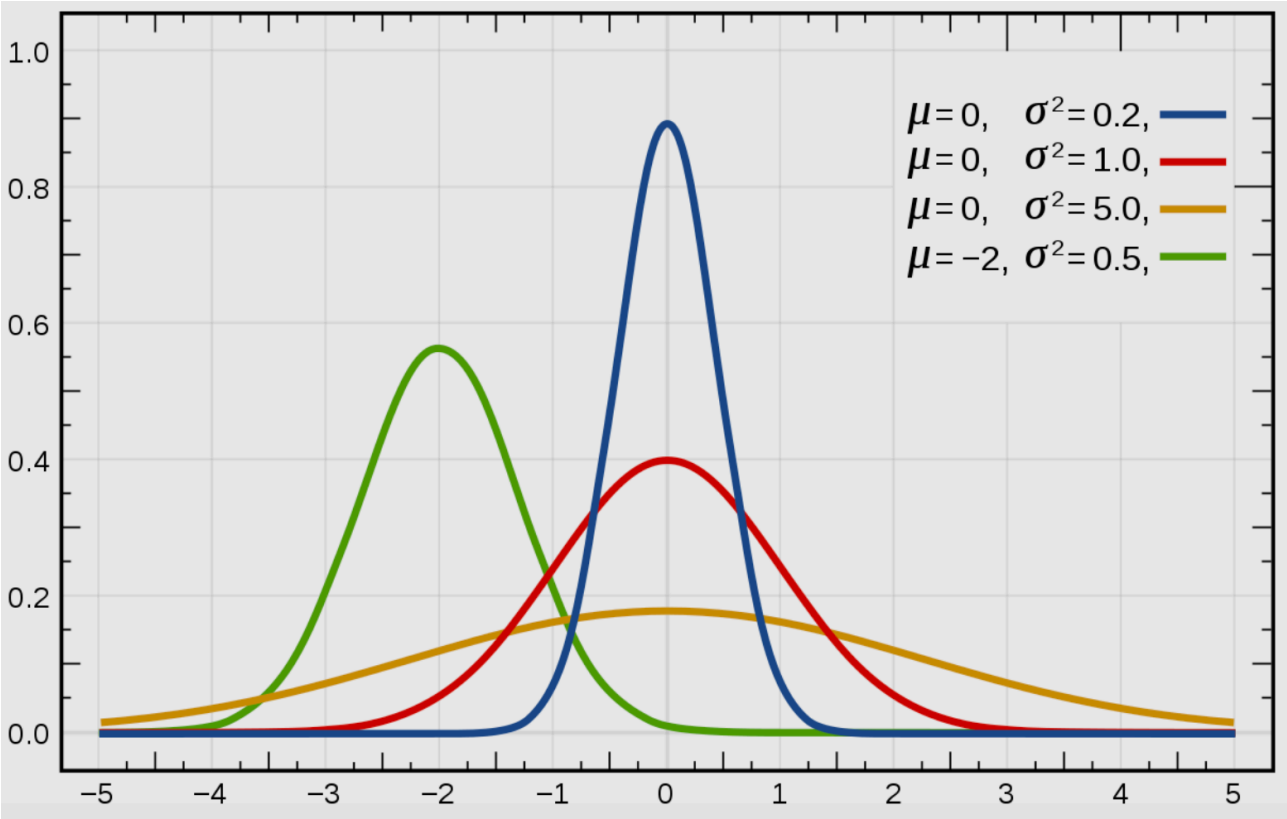
Our Focus: Parameterized Statistical Inference

- Given a parametric distribution model (a family of PMFs or PDFs) $\mathcal{F} = \{p(x;\theta) : \theta \in \mathcal{R}\}$
- θ is an unknown parameter in a parameter space $\mathcal{R}$
- Now given (random)sample from such model: $\boldsymbol{X} = (X_1, \ldots, X_n)$
- How to make parameterized statistical inference?

Example: Parameterized Distribution Model

$$\left\{ p(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2 \right), \mu \in \mathbb{R}, \sigma > 0 \right\}.$$

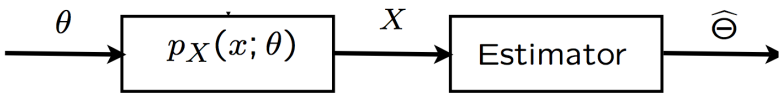


Parameterized Statistical Inference: Bayesian versus Frequentist

- Difference relates to the nature of the unknown parameter θ
- Treated as an unknown constant θ: **frequentist approach**
- Treated as a random variable Θ with prior (known) distribution: **Bayesian approach**

Statistical Inference: Frequentist Perspective

- Classical statistics: unknown constant θ



- Hypothesis testing: $H_0 : \theta = 1/2$ versus $H_1 : \theta = 3/4$
- Composite hypotheses: $H_0 : \theta = 1/2$ versus $H_1 : \theta \neq 1/2$
- Estimation: design an **estimator** $\widehat{\Theta}$, to "keep estimation **error** $\widehat{\Theta} - \theta$ small"

Estimation Method: Maximum Likelihood Estimation (MLE)

- We observe a particular data $\boldsymbol{x} = (x_1, \ldots, x_n)$,
- Likelihood: the probability (or probability density) of seeing data $\boldsymbol{x}$ under different values of parameter θ, i.e., $p(\boldsymbol{x}; \theta)$
- A **maximum likelihood estimate** (MLE) is a value of the parameter θ that maximizes the likelihood $p(\boldsymbol{x}; \theta)$ over all possible values:

$$\hat{\theta} = \arg\max_{\theta} p(\boldsymbol{x}; \theta)$$

MLE under Independent Case

- Random Sample: $\{X_i\}$ are i.i.d., we have

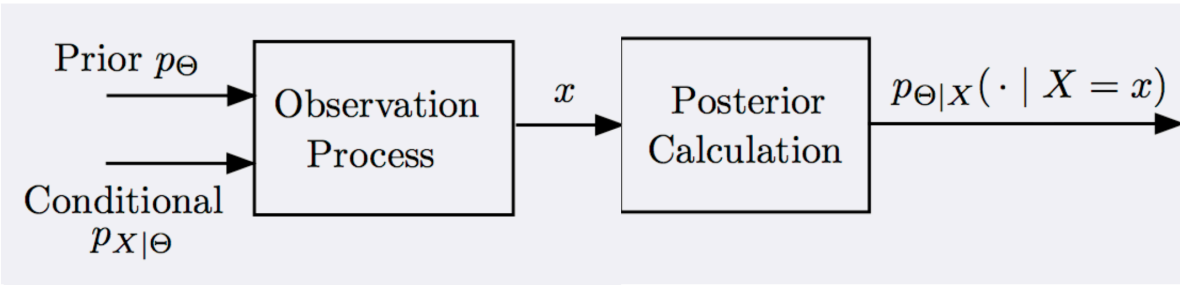
$$\log[p(\boldsymbol{x};\theta)] = \log \prod_{i=1}^{n} p(x_i;\theta) = \sum_{i=1}^{n} \log[p(x_i;\theta)]$$

- Thus a **maximum likelihood estimate** (MLE) under independent case is shown as follows:

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta}\ p(\boldsymbol{x};\theta) = \arg\max_{\theta}\ \log[p(\boldsymbol{x};\theta)] \\ &= \arg\max_{\theta}\ \sum_{i=1}^{n} \log[p(x_i;\theta)]\end{aligned}$$

Bayesian Inference

- Unknown Θ
  - treated as a random variable
  - prior distribution p_Θ or f_Θ
- Observation X
  - observation model $p_{X|\Theta}$ or $f_{X|\Theta}$
- Use appropriate version of the Bayes rule to find $p_{\Theta|X}(\cdot \mid X = x)$ or $f_{\Theta|X}(\cdot \mid X = x)$

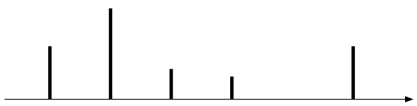


- Where does the prior come from?
  - symmetry
  - known range
  - earlier studies
  - subjective or arbitrary

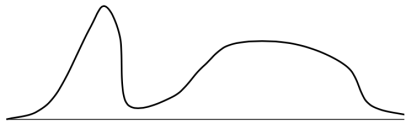
The Output of Bayesian Statistical Inference

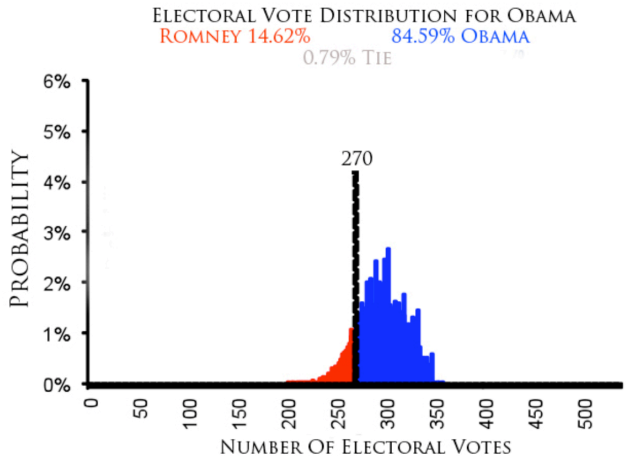
The complete answer is a posterior distribution:
PMF $p_{\Theta|X}(\cdot \mid x)$ or PDF $f_{\Theta|X}(\cdot \mid x)$

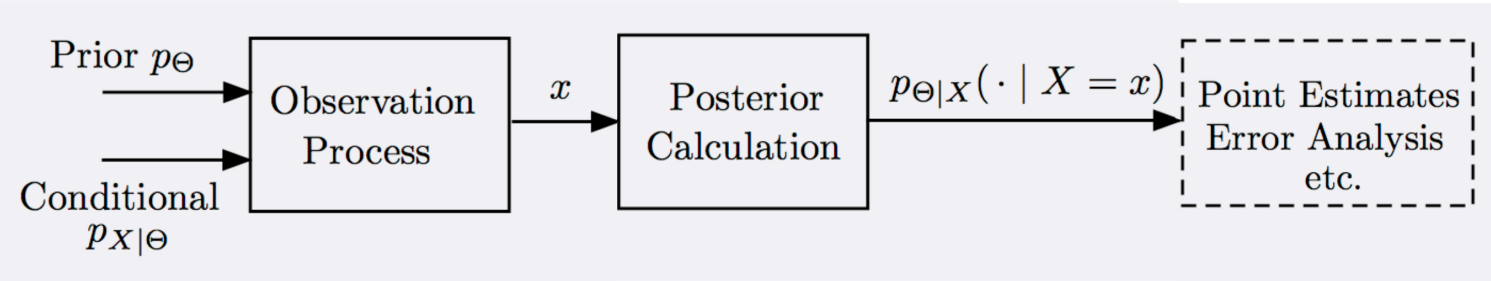
$p_{\Theta|X}(\cdot \mid x)$



$f_{\Theta|X}(\cdot \mid x)$







General Bayes' Rule for Bayesian Statistical Inference

The Four Versions of Bayes' Rule

- Θ discrete, X discrete:

$$p_{\Theta|X}(\theta \,|\, x) = \frac{p_\Theta(\theta) p_{X|\Theta}(x \,|\, \theta)}{\sum_{\theta'} p_\Theta(\theta') p_{X|\Theta}(x \,|\, \theta')}.$$

- Θ discrete, X continuous:

$$p_{\Theta|X}(\theta \,|\, x) = \frac{p_\Theta(\theta) f_{X|\Theta}(x \,|\, \theta)}{\sum_{\theta'} p_\Theta(\theta') f_{X|\Theta}(x \,|\, \theta')}.$$

- Θ continuous, X discrete:

$$f_{\Theta|X}(\theta \,|\, x) = \frac{f_\Theta(\theta) p_{X|\Theta}(x \,|\, \theta)}{\int f_\Theta(\theta') p_{X|\Theta}(x \,|\, \theta') \, d\theta'}.$$

- Θ continuous, X continuous:

$$f_{\Theta|X}(\theta \,|\, x) = \frac{f_\Theta(\theta) f_{X|\Theta}(x \,|\, \theta)}{\int f_\Theta(\theta') f_{X|\Theta}(x \,|\, \theta') \, d\theta'}.$$

Bayes' Rule: Bayesian Perspective

Theorem

A represents cognition, B represents data, then

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

- $P(A)$: *Prior Cognition*
- $P(B)$: *Evidence*
- $P(B|A)$: *Likelihood (info of data given prior cognition)*
- $P(A|B)$: *Posterior Cognition*

Estimation Method I

- The Maximum A Posteriori Probability (MAP)
- Given the observation value x, the MAP rule selects a value $\hat{\theta}$ that maximizes the posterior probability(probability density) $p_{\Theta|\boldsymbol{X}}(\theta|\boldsymbol{x})$:

$$\hat{\theta} = \arg\max_{\theta} p_{\Theta|\boldsymbol{X}}(\theta|\boldsymbol{x})$$

- Equivalently,

$$\hat{\theta} = \arg\max_{\theta} p_{\Theta}(\theta) p_{\boldsymbol{X}|\Theta}(\boldsymbol{x}|\theta)$$

Estimation Method II

- Posterior Mean: Given the observation data $\boldsymbol{x}$, the estimate of Θ is $\hat{\theta} = E[\Theta|\boldsymbol{X} = \boldsymbol{x}]$
- Now given the observation sample $\boldsymbol{X}$, the estimator of Θ is $\hat{\Theta} = E[\Theta|\boldsymbol{X}]$

Conjugate Prior

- Before Monte Carlo, posterior calculation is hard
- Conjugate Prior: reduce the computing complexity of posterior distribution
- Loosely speaking, a prior distribution is conjugate to the likelihood model if both the prior and posterior distribution stay in the same distribution family.

Story: Beta-Binomial Conjugacy

We have a coin that lands Heads with probability θ , but we don't know what θ is. Our goal is to infer the value of θ after observing the outcomes of n tosses of the coin. The larger that n is, the more accurately we should be able to estimate p.

Bayesian Inference

- Treats all unknown quantities as random variables.
- In the Bayesian approach, we would treat the unknown probability θ as a random variable and give θ a distribution.
- This is called a **prior distribution**, and it reflects our uncertainty about the true value of θ before observing the coin tosses.
- After the experiment is performed and the data are gathered, the prior distribution is updated using Bayes' rule; this yields the **posterior distribution**, which reflects our new beliefs about θ.
- Now we adopt the beta distribution as the prior distribution.

Beta Distribution

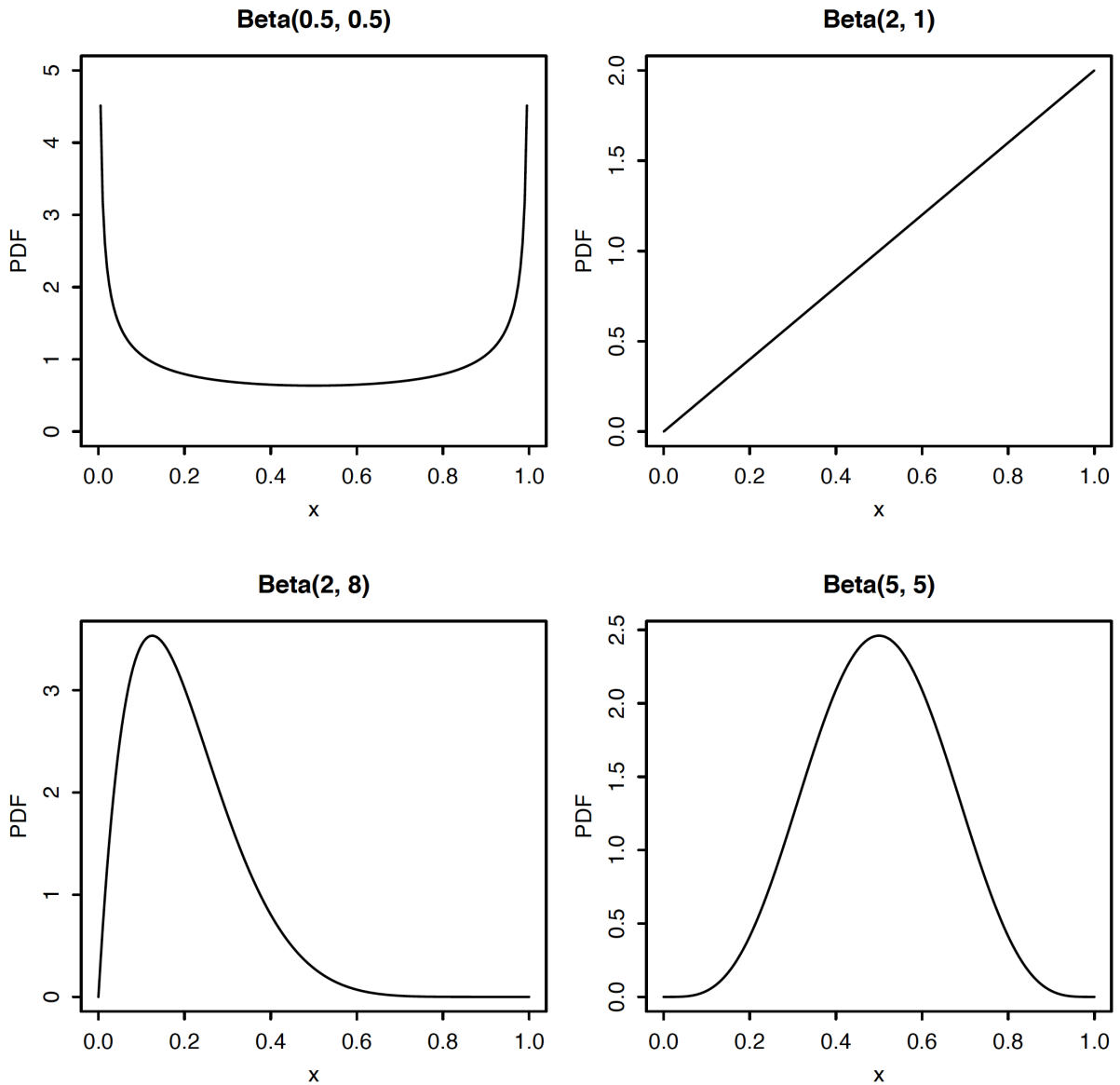
Definition

An r.v. X is said to have the *Beta distribution* with parameters a and b, $a > 0$ and $b > 0$, if its PDF is

$$f(x) = \frac{1}{\beta(a,b)} x^{a-1}(1-x)^{b-1}, \ 0 < x < 1,$$

where the constant $\beta(a,b)$ is chosen to make the PDF integrate to 1. It is denoted as $X \sim \mathrm{Beta}(a,b)$. Beta distribution is a generalization of uniform distribution.

PDF of Beta Distribution



Story: Beta-Binomial Conjugacy

Story: Beta-Binomial Conjugacy

Story: Beta-Binomial Conjugacy

- Furthermore, notice the very simple formula for updating the distribution of θ.
- We just add the number of observed successes, k, to the first parameter of the Beta distribution.
- We also add the number of observed failures, $n - k$, to the second parameter of the Beta distribution.
- So a and b have a concrete interpretation in this context:
  - a as the number of prior successes in earlier experiments
  - b as the number of prior failures in earlier experiments
  - a, b: pseudo counts

Mean vs. Bayesian(Posterior) Average

- Infer the value of θ (probability of coin lands heads)
- Observed k heads out of n tosses of the coin
- Mean: $\frac{k}{n}$
- Bayesian(Posterior) Average: $\mathbb{E}(\theta|X=k)=\frac{a+k}{a+b+n}$
- Suppose the prior distribution is Unif(0,1): $a=1, b=1$
- Bayesian(Posterior) Average: $\frac{k+1}{n+2}$
- When $k=n$, we have: 1 (mean) vs. $\frac{n+1}{n+2}$ (Bayesian average)

Bayesian(Posterior) Average

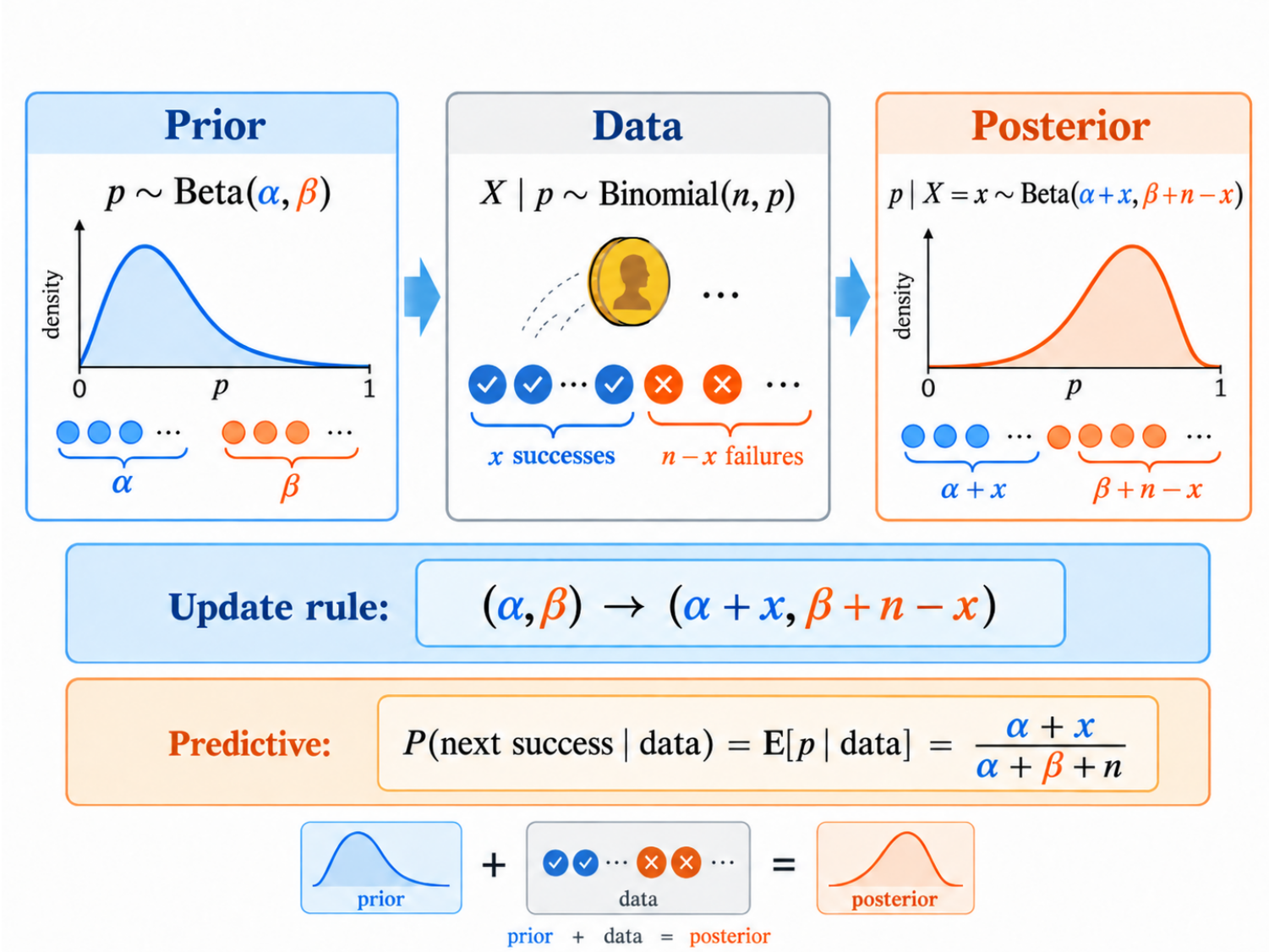
- Prior Average: $\mathbb{E}(\theta) = \frac{a}{a+b}$
- Bayesian(Posterior) Average: $\mathbb{E}(\theta|X = k) = \frac{a+k}{a+b+n}$

$$\mathbb{E}[\theta \mid X = x] = \underbrace{\frac{a+b}{a+b+n}}_{\text{先验权重}} \cdot \underbrace{\frac{a}{a+b}}_{\text{先验均值}} + \underbrace{\frac{n}{a+b+n}}_{\text{数据权重}} \cdot \underbrace{\frac{k}{n}}_{\text{样本成功比例}}$$

Story: Beta-Binomial Conjugacy

If we have a Beta prior distribution on θ and data that are conditionally Binomial given θ, then when going from prior to posterior, we don't leave the family of Beta distributions. We say that **the Beta distribution is the conjugate prior of the Binomial distribution**.

Story: Beta-Binomial Conjugacy



Likelihood Model: Discrete

Sample Space	Sampling Dist.	Conjugate Prior	Posterior
$\mathcal{X} = \{0, 1\}$	Bernoulli(θ)	Beta(α, β)	Beta$(\alpha + n\overline{X}, \beta + n(1 - \overline{X}))$
$\mathcal{X} = \mathbb{Z}_+$	Poisson(λ)	Gamma(α, β)	Gamma$(\alpha + n\overline{X}, \beta + n)$
$\mathcal{X} = \mathbb{Z}_{++}$	Geometric(θ)	Gamma(α, β)	Gamma$(\alpha + n, \beta + n\overline{X})$
$\mathcal{X} = \mathbb{H}_K$	Multinomial(θ)	Dirichlet(α)	Dirichlet$(\alpha + n\overline{X})$

Likelihood Model: Continuous

Sampling Dist.	Conjugate Prior	Posterior
Uniform(θ)	Pareto(ν_0, k)	Pareto $\left(\max\{\nu_0, X_{(n)}\}, n+k\right)$
Exponential(θ)	Gamma(α, β)	Gamma$(\alpha+n, \beta+n\overline{X})$
$N(\mu, \sigma^2)$, known σ^2	$N(\mu_0, \sigma_0^2)$	$N\left(\left(\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}\right)^{-1}\left(\frac{\mu_0}{\sigma_0^2}+\frac{n\overline{X}}{\sigma^2}\right), \left(\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}\right)^{-1}\right)$
$N(\mu, \sigma^2)$, known μ	InvGamma(α, β)	InvGamma $\left(\alpha+\frac{n}{2}, \beta+\frac{n}{2}\overline{(X-\mu)^2}\right)$
$N(\mu, \sigma^2)$, known μ	ScaledInv-$\chi^2(\nu_0, \sigma_0^2)$	ScaledInv-$\chi^2\left(\nu_0+n, \frac{\nu_0\sigma_0^2}{\nu_0+n}+\frac{n\overline{(X-\mu)^2}}{\nu_0+n}\right)$
$N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, known $\boldsymbol{\Sigma}$	$N(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$	$N\left(\mathbf{K}\left(\boldsymbol{\Sigma}_0^{-1}\mu_0+n\boldsymbol{\Sigma}^{-1}\overline{X}\right), \mathbf{K}\right)$, $\mathbf{K}=\left(\boldsymbol{\Sigma}_0^{-1}+n\boldsymbol{\Sigma}^{-1}\right)^{-1}$
$N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, known $\boldsymbol{\mu}$	InvWishart$(\nu_0, \mathbf{S}_0)$	InvWishart$(\nu_0+n, \mathbf{S}_0+n\overline{\mathbf{S}})$, $\overline{\mathbf{S}}$ sample covariance

Outline

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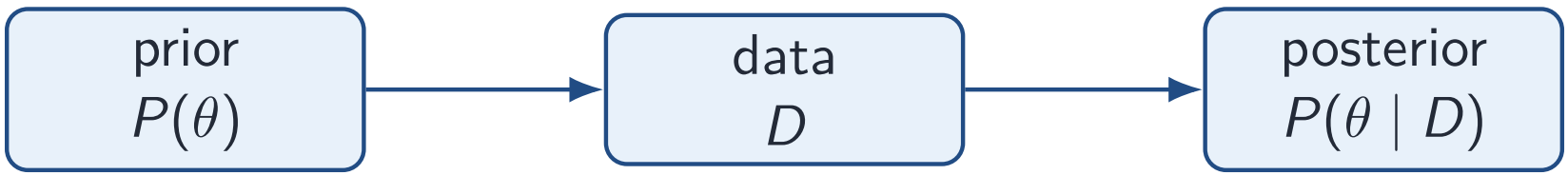
2. Expectation & Conditional Expectation
3. Conditional Expectation as Projection Geometry
4. Concentration Inequality
5. Markov Dynamics I
6. Markov Dynamics II
7. Statistical Inference
8. Bayesian Learning: RL Perspective
9. Summary

Bayesian Learning as Belief Update

Bayes rule

$$P(\theta \mid D) = \frac{P(D \mid \theta)P(\theta)}{P(D)}$$



Core idea

Bayesian learning keeps uncertainty instead of only a point estimate.

Bayes Rule in 2–3 Steps

Start from the product rule:

$$P(\theta, D) = P(D \mid \theta)P(\theta).$$

Also,

$$P(\theta, D) = P(\theta \mid D)P(D).$$

Equate and solve:

$$\boxed{P(\theta \mid D) = \frac{P(D \mid \theta)P(\theta)}{P(D)}.}$$

RL reading

Belief updates in POMDPs are Bayes updates over hidden states.

Beta-Bernoulli Example

Unknown success probability θ of an arm.

Prior

$$\theta \sim \mathrm{Beta}(\alpha, \beta).$$

Data

Observe s successes and f failures.

Posterior

$$\boxed{\theta \mid D \sim \mathrm{Beta}(\alpha + s, \beta + f).}$$

Posterior mean

$$\mathbb{E}[\theta \mid D] = \frac{\alpha + s}{\alpha + \beta + s + f}.$$

Small Example: Bayesian Update

Prior:

$$\theta \sim \mathrm{Beta}(1, 1).$$

Data: 7 successes and 3 failures.

Posterior

$$\theta \mid D \sim \mathrm{Beta}(8, 4).$$

Posterior mean

$$\mathbb{E}[\theta \mid D] = \frac{8}{12} = 0.667.$$

Bandit preview

Thompson sampling draws a plausible θ from the posterior and acts greedily with respect to that draw.

Posterior Predictive Distribution

Bayesian learning predicts future observations by averaging over parameter uncertainty:

$$P(x_{new} \mid D) = \int P(x_{new} \mid \theta, D) P(\theta \mid D) d\theta.$$

Beta-Bernoulli case

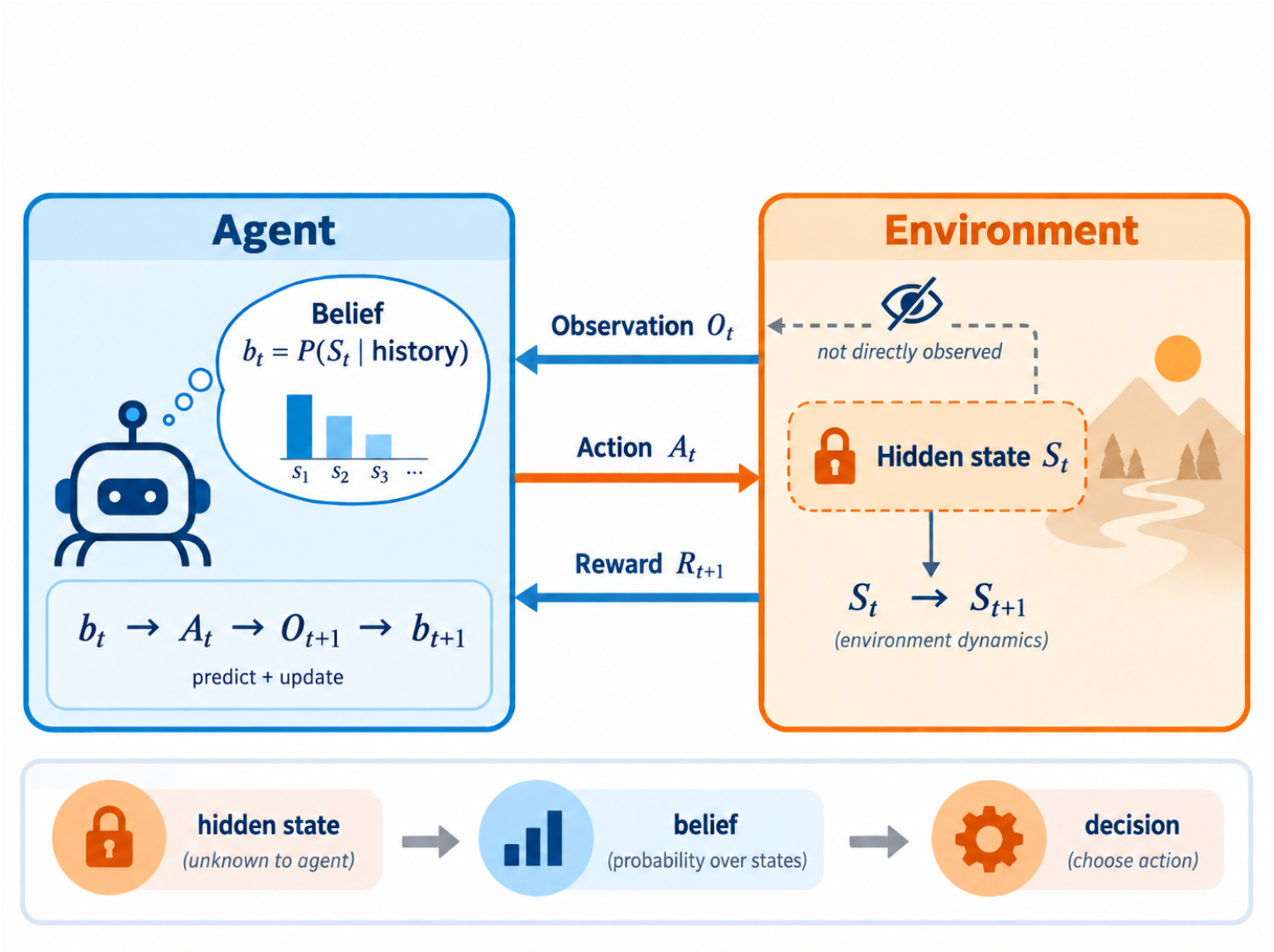
The probability of success next time is

$$P(X_{new} = 1 \mid D) = \mathbb{E}[\theta \mid D] = \frac{\alpha + s}{\alpha + \beta + s + f}.$$

Core idea

Bayesian prediction integrates uncertainty instead of plugging in only one estimate.

POMDP: Partially Observable Markov Decision Process



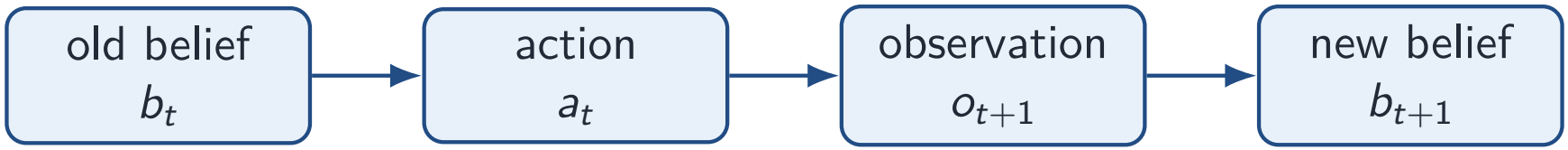
Belief State in a POMDP Preview

Hidden state

The agent does not observe S_t directly. It observes O_t.

Belief

$$b_t(s) = P(S_t = s \mid O_{1:t}, A_{1:t-1}).$$



One-Step Belief Update

Prediction step:

$$\bar{b}_{t+1}(s') = \sum_s P(s' \mid s, a_t) b_t(s).$$

Correction step:

$$b_{t+1}(s') = \frac{P(o_{t+1} \mid s') \bar{b}_{t+1}(s')}{\sum_x P(o_{t+1} \mid x) \bar{b}_{t+1}(x)}.$$

Key idea

Markov prediction + Bayesian correction.

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Lecture 1 Checklist

You should now be able to explain:

1. why value functions are expectations;
2. how Monte Carlo estimation works;
3. why uncertainty shrinks as sample size grows;
4. how Hoeffding gives a confidence interval;
5. how a transition matrix evolves a distribution;
6. how to solve $\pi = \pi Q$ in a small chain;
7. what mixing means;
8. how Bayes rule updates beliefs.

Suggested Self-Test Exercises

1. Estimate a reward mean from 10 samples and compute a Hoeffding confidence radius.
2. Given a two-state transition matrix, compute μ_1, μ_2, and the stationary distribution.
3. Compare two chains: which one mixes faster and why?
4. Perform a Beta-Bernoulli posterior update after observing successes and failures.
5. Interpret a belief update as "prediction + correction".

Pedagogical goal

No long proofs this week. The goal is to build the mathematical reflexes used throughout RL.

Takeaway from Lecture 1

Probability tells us what value means.

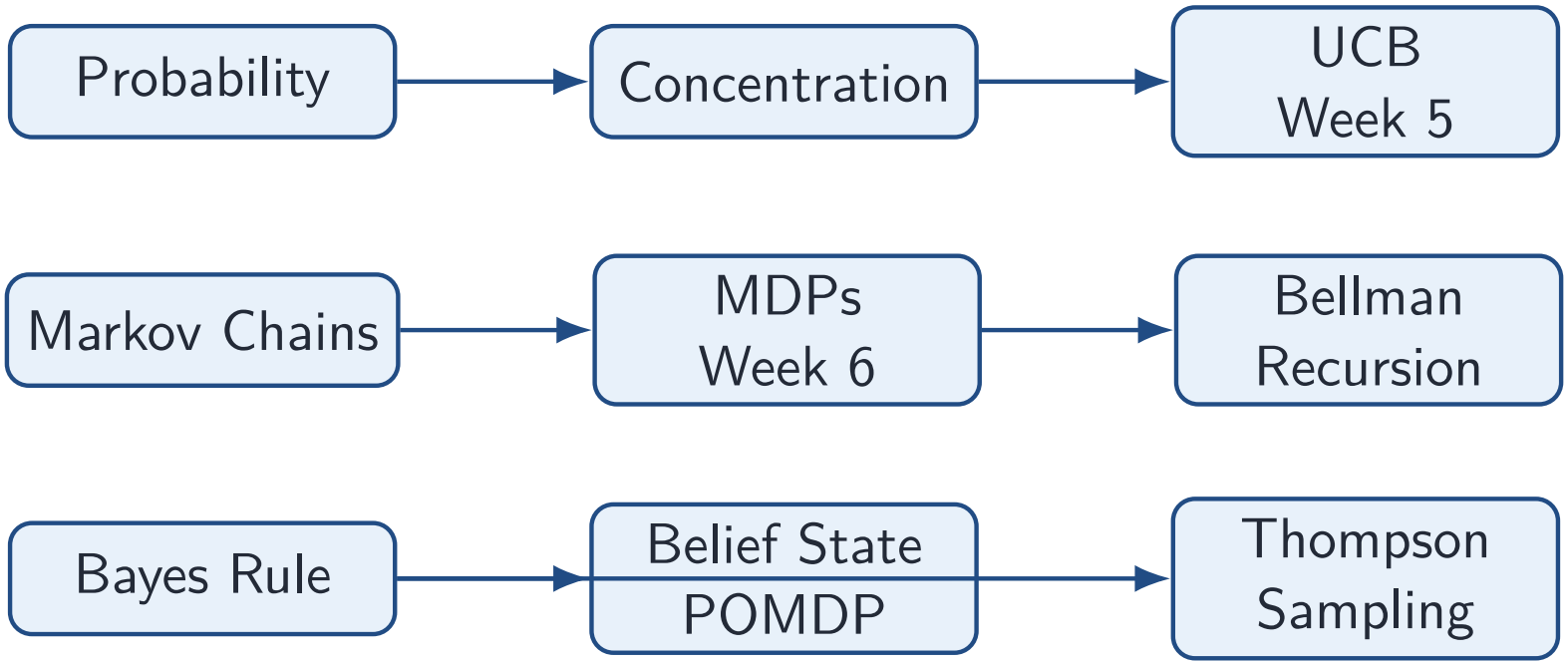
Concentration tells us when to trust estimates.

Markov dynamics tells us how states evolve.

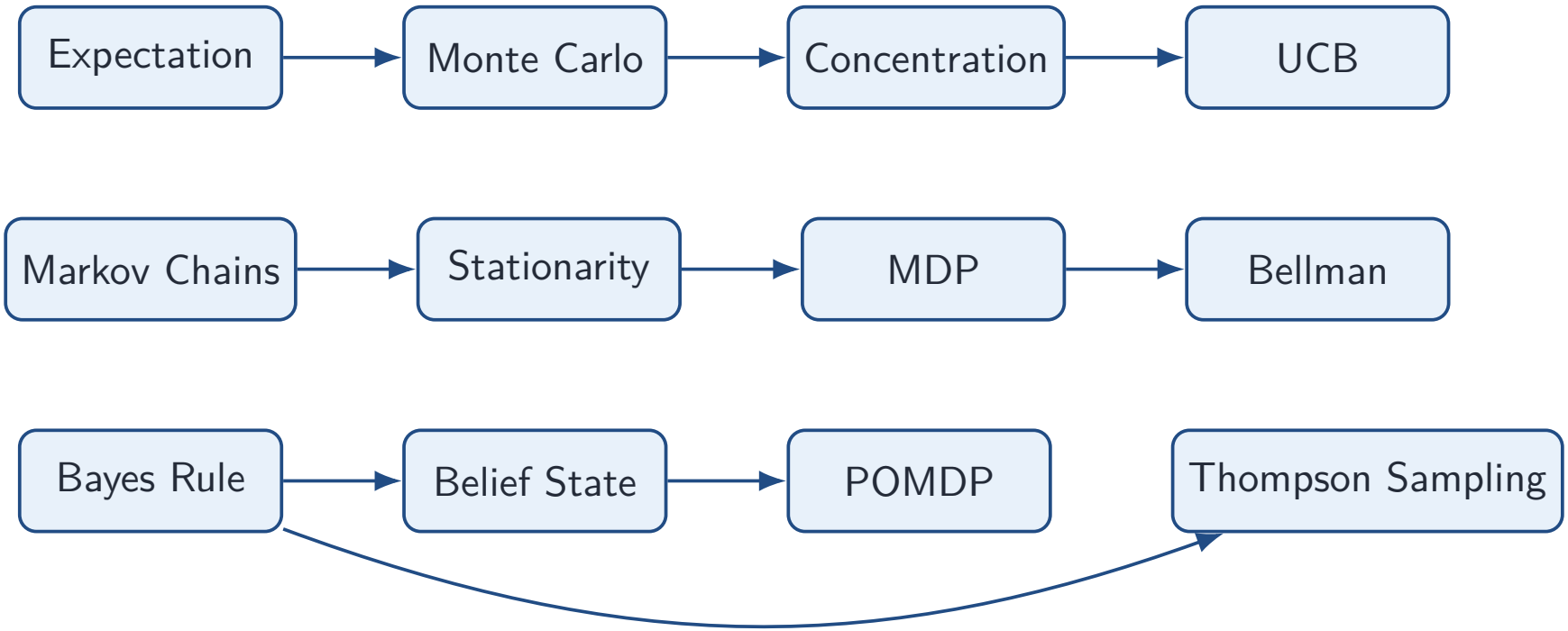
Bayesian inference tells us how beliefs change.

These are the foundations of reinforcement learning.

From Week 1 to the Rest of the Course



Course-Level Synthesis



Reading and Preparation

Before Lecture 2

Review:

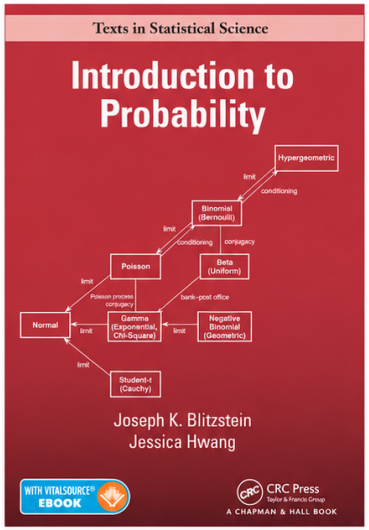
- expectation and conditional expectation;
- empirical mean and confidence intervals;
- basic Markov chain matrix multiplication;
- Bayes rule and posterior predictive probability.

Bridge to Lecture 2

When direct computing of expectation is hard, we need better sampling.

$$\text{Monte Carlo} \to \text{Importance Sampling} \to \text{MCMC} \to \text{Diffusion/Flow}.$$

Reference


Texts in Statistical Science
Introduction to Probability
Hypergeometric
Binomial (Bernoulli)
Poisson
Beta (Uniform)
Normal
Gamma (Exponential, Chi-Square)
Negative Binomial (Geometric)
Student-t (Cauchy)
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conditioning
conjugacy
Poisson process conjugacy
bank–post office
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Jessica Hwang
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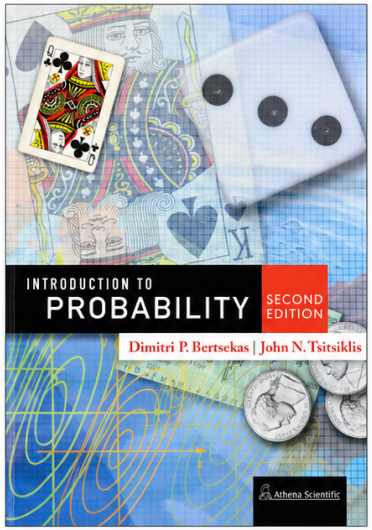
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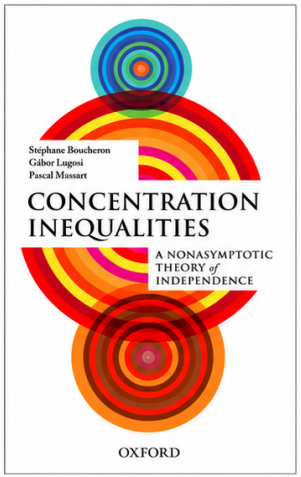
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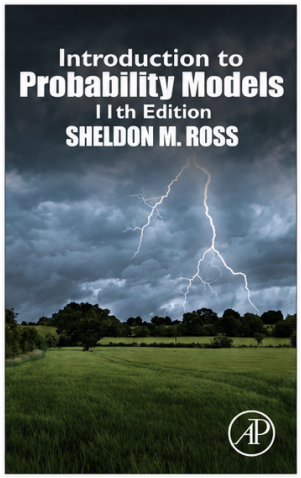
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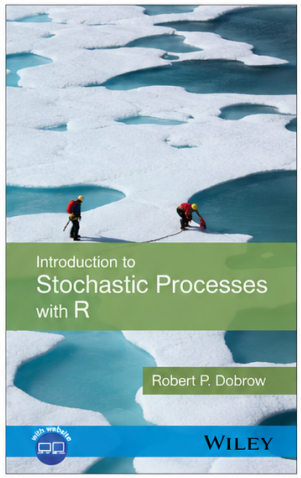
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